

Lecture 09: Deep Wireless Sensing

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THE UNIVERSITY OF HONG KONG



Contents

- Framework
- Signal Representation
- Data
 - (1) How do we augment data for wireless sensing?
 - (2) How do we generate synthesis data for wireless sensing?
 - (3) How do we build large-scale real-world dataset for wireless sensing?
- Model
 - (1) How do we customize deep neural network for wireless sensing?
 - (2) How do we combine SP and DL for wireless sensing?
 - (3) How do we ensure interpretability for deep wireless sensing?
 - (4) Building a Foundation Model for wireless sensing?

WiFi Sensing: A 10-year Journey

- SP-based: Geometrical Approaches (Reflection Model)
- SP-based: Statistical Approaches (Scattering Model)
- DL-based: Deep Wireless Sensing (Neural Network Model)
 - Wireless Data for Learning
 - Model Design for Wireless Data

Why (Not) Deep Learning?

- Why Not

- Efficient
- Explainable
- Deployable (on IoT)
- Data

- Why

- Enabling more applications that are difficult to achieve with SP alone



Deep Wireless Sensing

- We will use WiFi signals as example in this lecture
- mmWave and other RF signals are similar

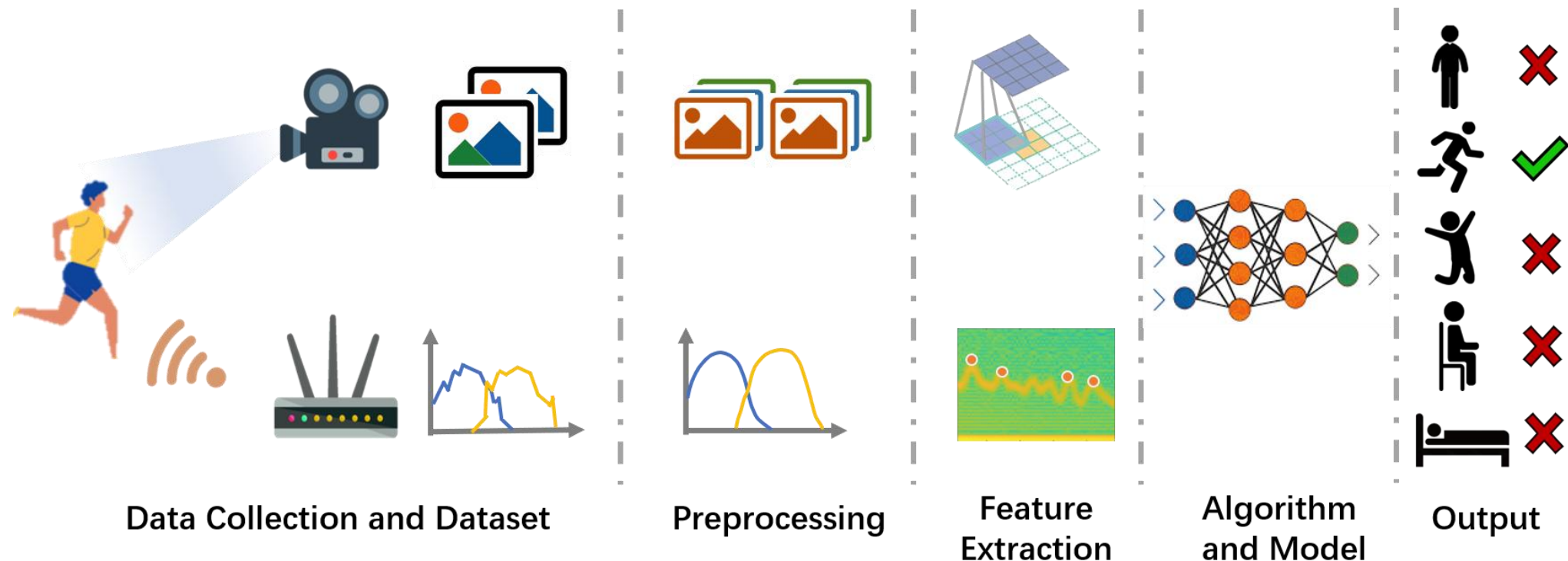


Diagram source: <http://tns.thss.tsinghua.edu.cn/wst/docs/intro>

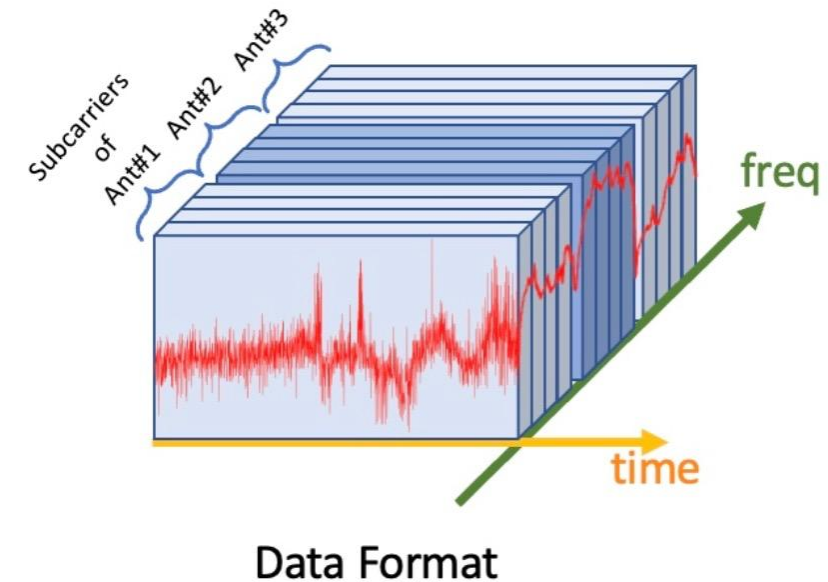
Questions to Ask

- Data: How can we collect (sufficient) data for training?
- Feature: What features should be used for learning?
- Model: What models should we use?
- *Deployment: How can the trained models be deployed?



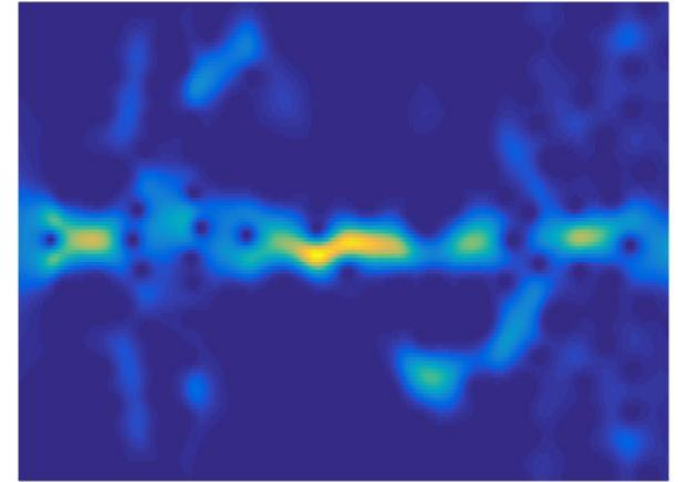
CSI Data for Learning

- **Non-visual**
 - Contain physical and geometric connotations in time, space, and frequency domains all non-visually intelligible (like images to human eyes)
 - (They can certainly be visualized)
 - Visual data are visual, of course
- **Complex**
 - Complex-valued tensors with amplitude and phase
 - Visual data are real-valued
- **High-dimensional**
 - time, subcarrier, antennas, transceivers
 - Visual data are usually 2D (image) or 3D (video)

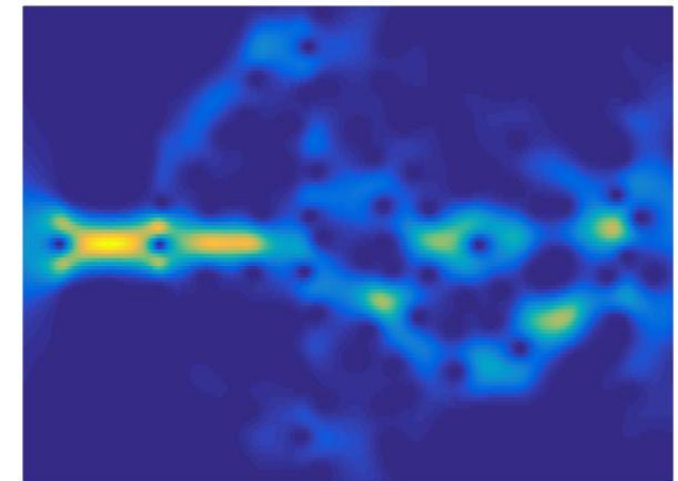


CSI Datasets

- More difficult to collect
 - Much more complex setups
 - Less reliable platforms for data collection
 - Not always capture valid data
 - Not all captured data are useful/meaningful
- Depends on many environmental factors
 - Users, device placements, user locations, user orientations, places/rooms, environmental factors like furniture, wireless configs, device heterogeneity...



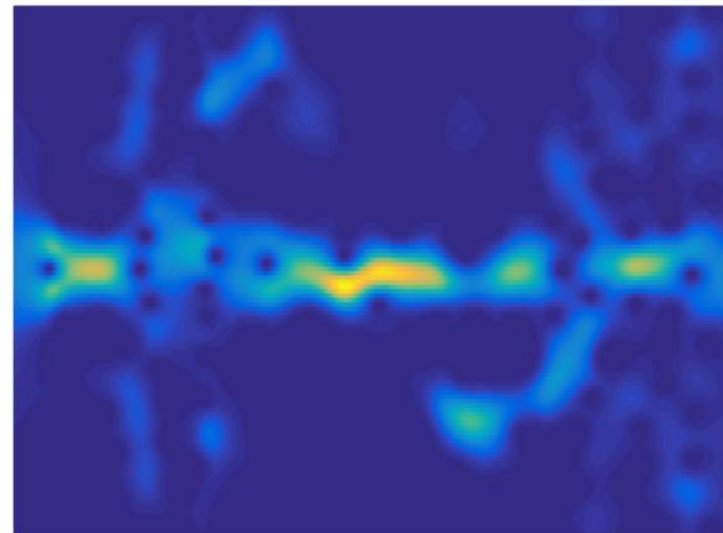
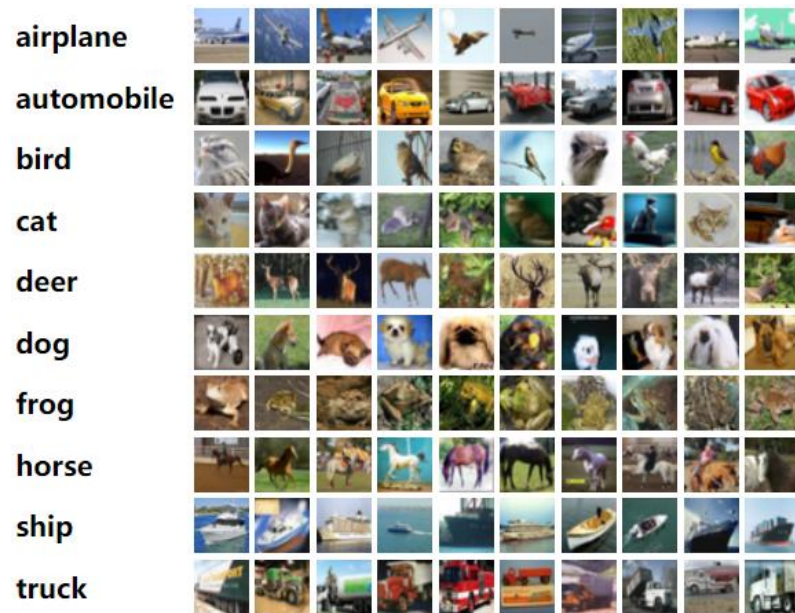
CSI in Setting 1



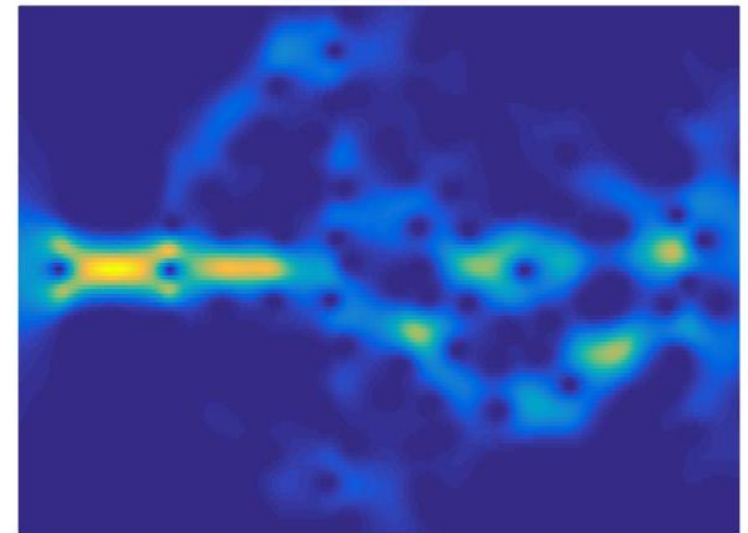
CSI in Setting 2

CSI Datasets

- More difficult to label
 - Cannot be labelled OFFLINE (unlike images!)



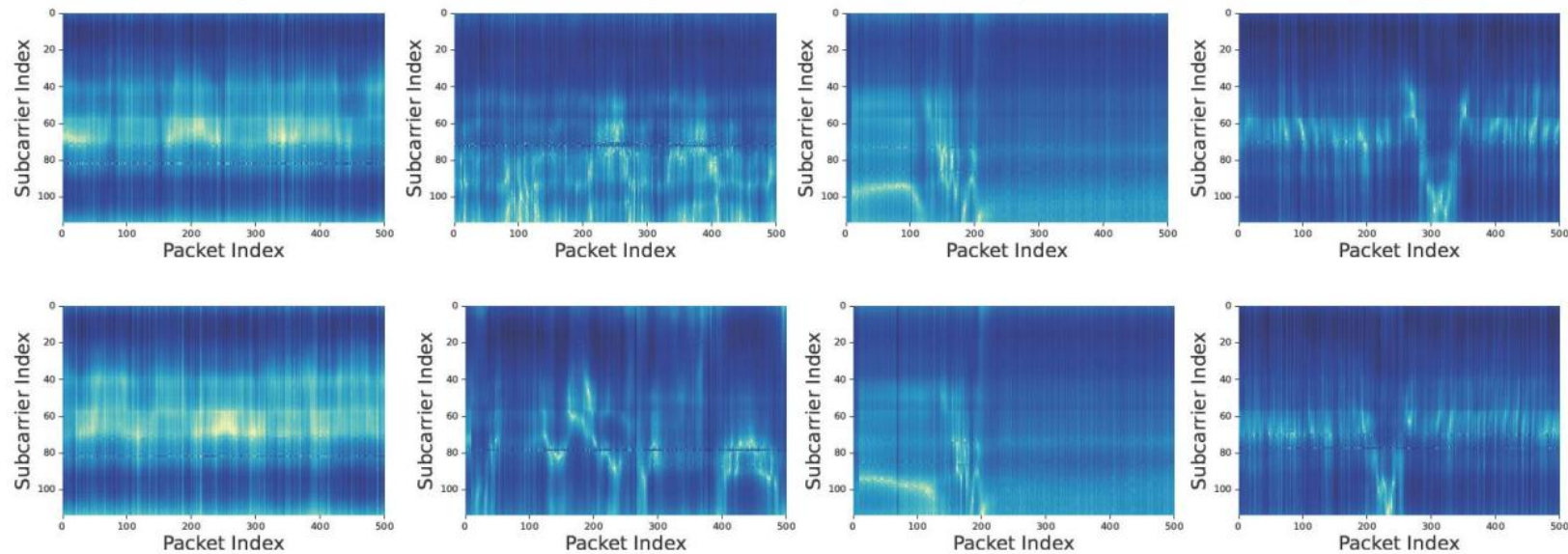
What is this?



And this??

CSI Datasets

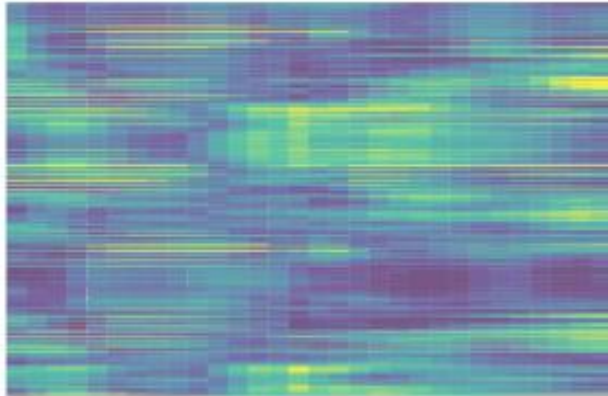
- More difficult to label
 - Cannot be labelled OFFLINE (unlike images!)



What are these?

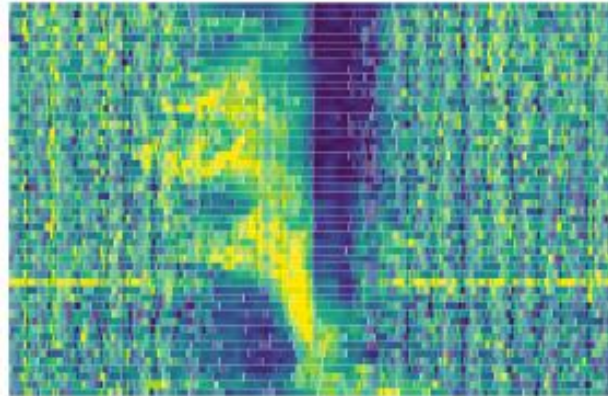
CSI Datasets

- More difficult to label
 - Cannot be labelled OFFLINE (unlike images!)



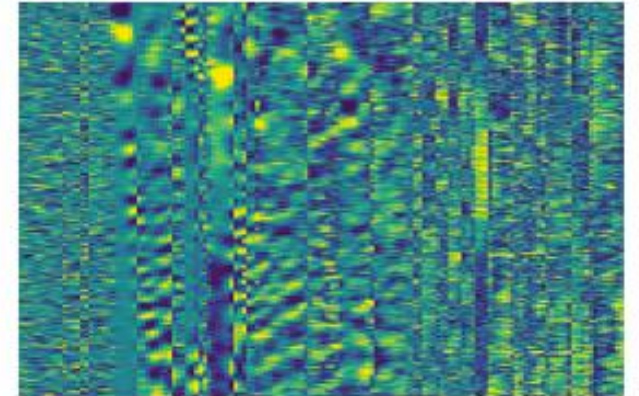
(b) Wi-Fi matrix.

What is this?



(c) FMCW matrix.

And this??

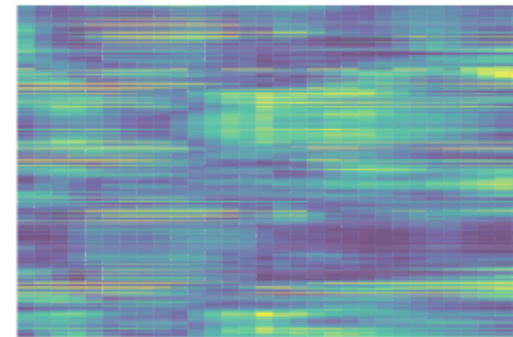


(d) IR matrix.

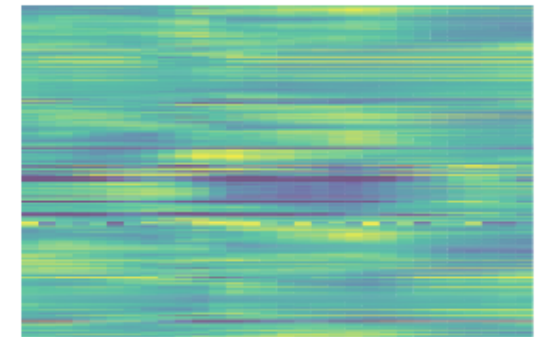
And this???

CSI Datasets

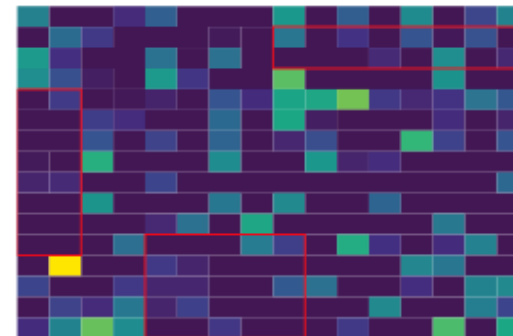
- Lack of large-scale datasets
 - **Widar3** dataset: 17 participants performing 22 gestures from 5 orientations towards one transmitter, standing at 5 different locations within the coverage of six receivers in 3 environments.
 - **300K** vs. **14M** (ImageNet)
- Difficult to learn
 - Too many impacting factors



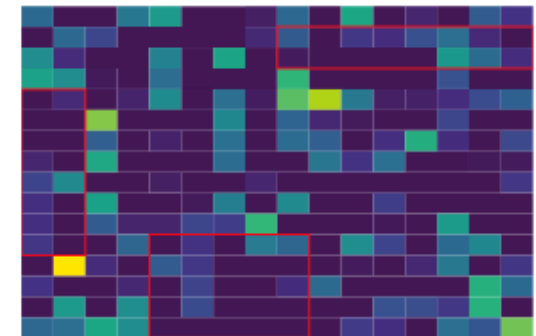
(a) Wi-Fi CSI in environment 1.



(b) Wi-Fi CSI in environment 2.

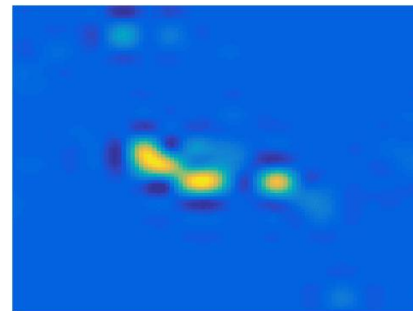
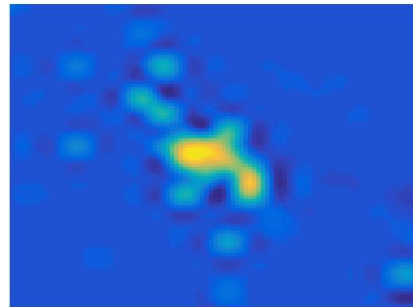
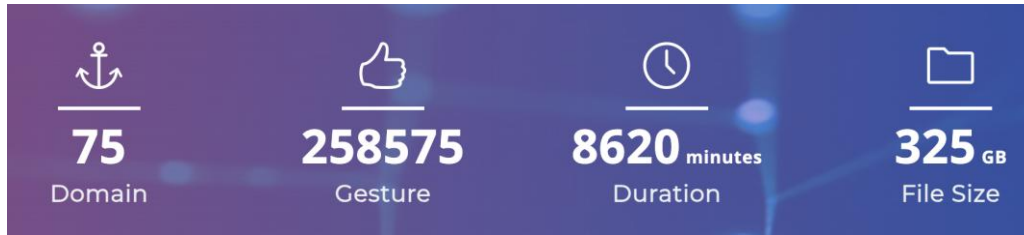


(c) Feature map of environment 1.



(d) Feature map of environment 2.

Widar3.0 dataset



IEEE DataPort™

DATASETS

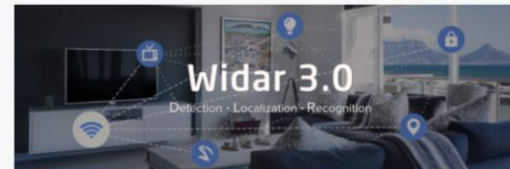
SUBMIT A DATASET

COMPETITIONS



Datasets

WIDAR 3.0: WIFI-BASED ACTIVITY RECOGNITION DATASET



Citation Author(s): Zheng Yang (School of Software, Tsinghua University)
Yi Zhang (School of Software, Tsinghua University)
Guidong Zhang (School of Software, Tsinghua University)
Yue Zheng (School of Software, Tsinghua University)
Guoxuan Chi (School of Software, Tsinghua University)

Submitted by: ZHENG YANG

Last updated: Wed, 03/06/2024 - 08:20

DOI: 10.21227/7znf-qp86

Links: Widar 3.0 Homepage
Professor Zheng Yang's Homepage

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<https://ieee-dataport.org/open-access/widar-30-wifi-based-activity-recognition-dataset>

From SP to DL

- How to perform signal processing so that the processed data would produce better learning performance?
- How to design neural networks so that they will better fit the unique wireless data (that are very different from visual data)?

Model-based
Signal Processing

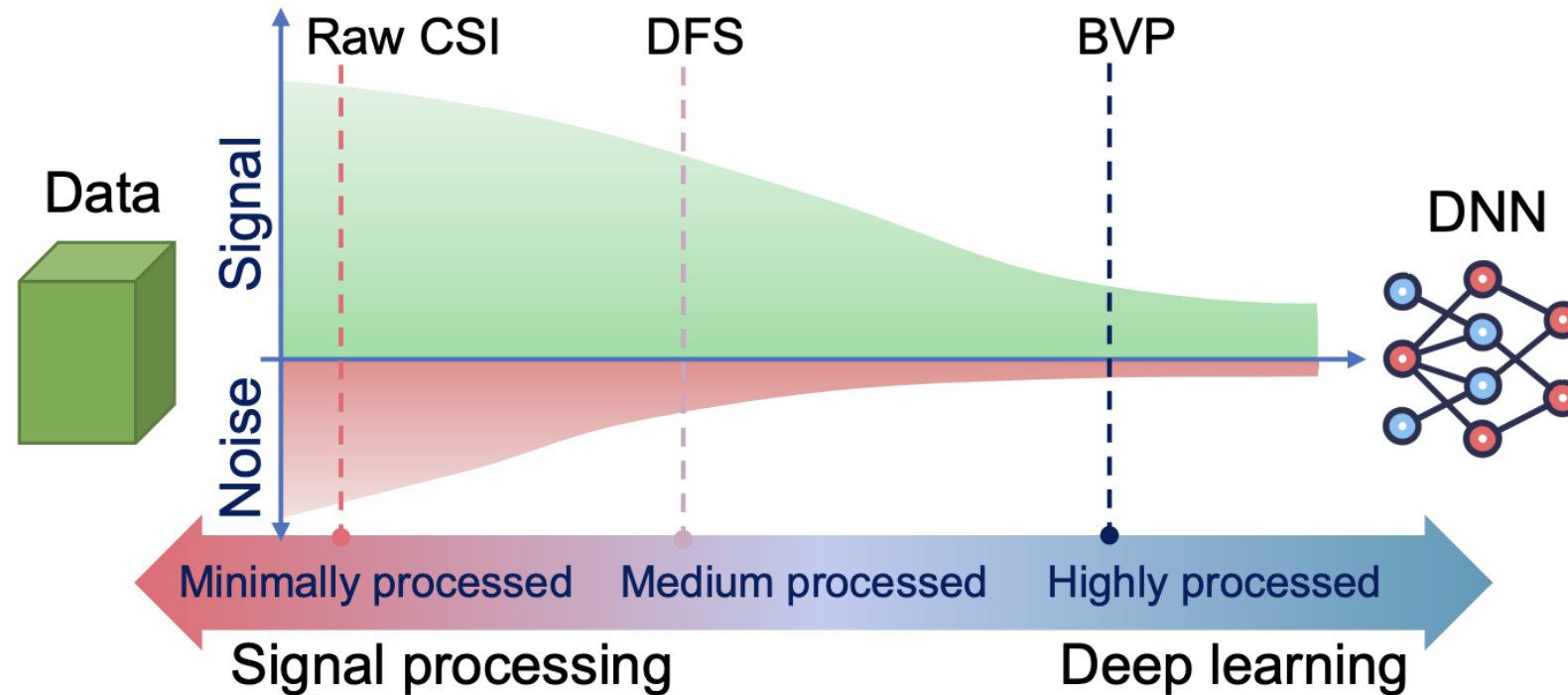


Data-driven
Deep Learning

model-guided,
data-driven
design?

Signal processing vs. Deep learning

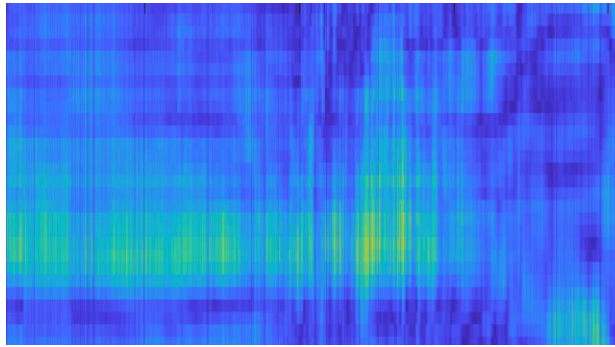
- A “sweet point” of balancing SP and DL?



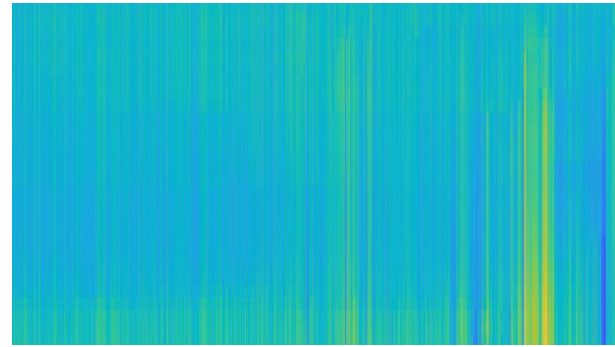
*BVP: Body-coordinate Velocity Profiles (BVPs) proposed in Widar3

Wireless Data Representation

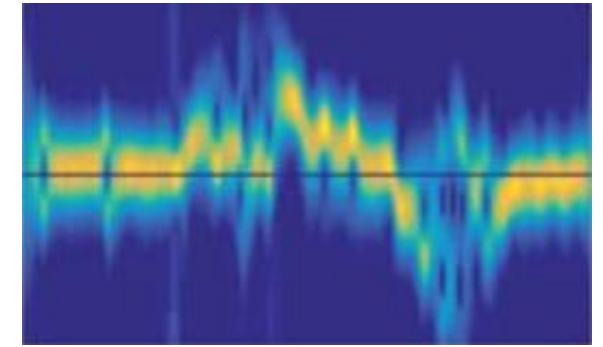
- WiFi CSI



Amplitude



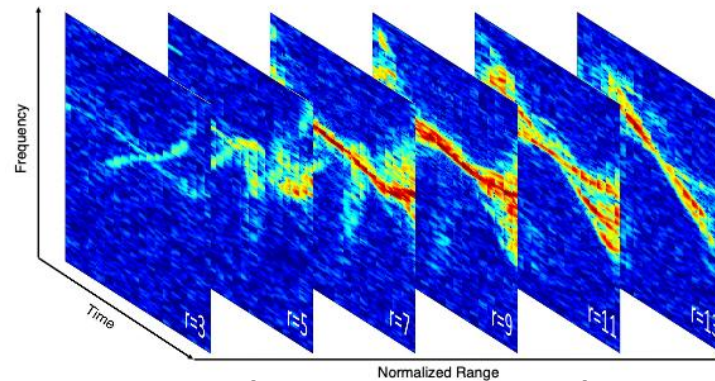
Phase



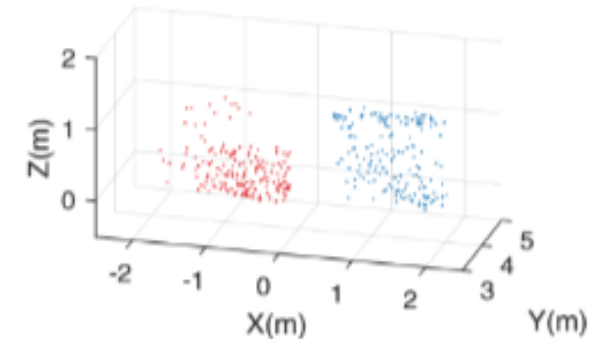
Spectrogram

...

- mmWave radar data



Doppler-Range Cube

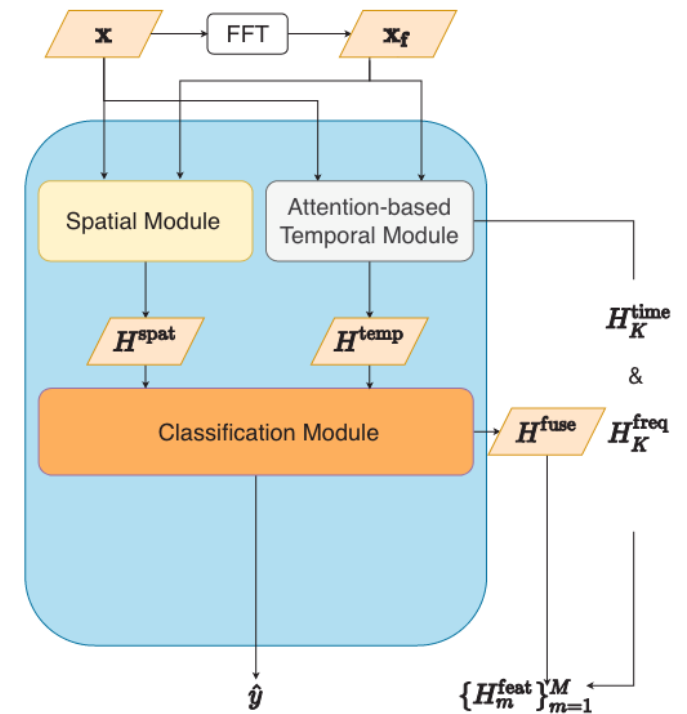
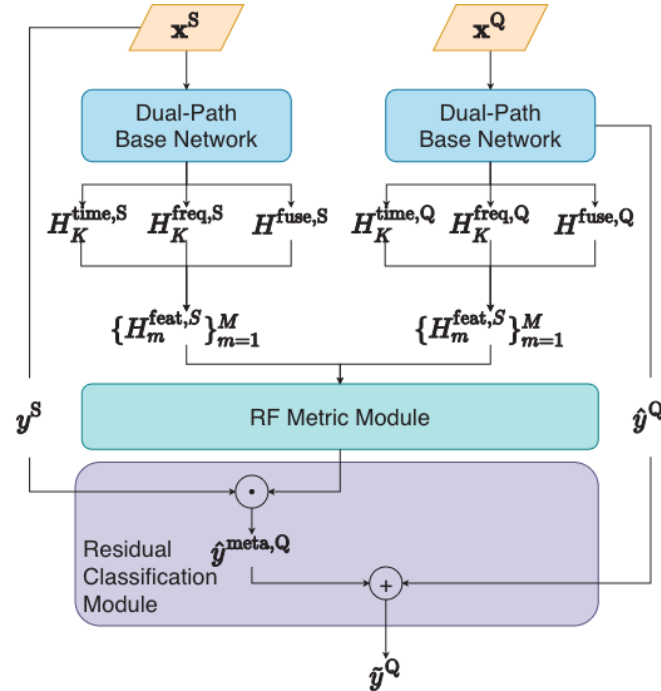


Point Clouds

...

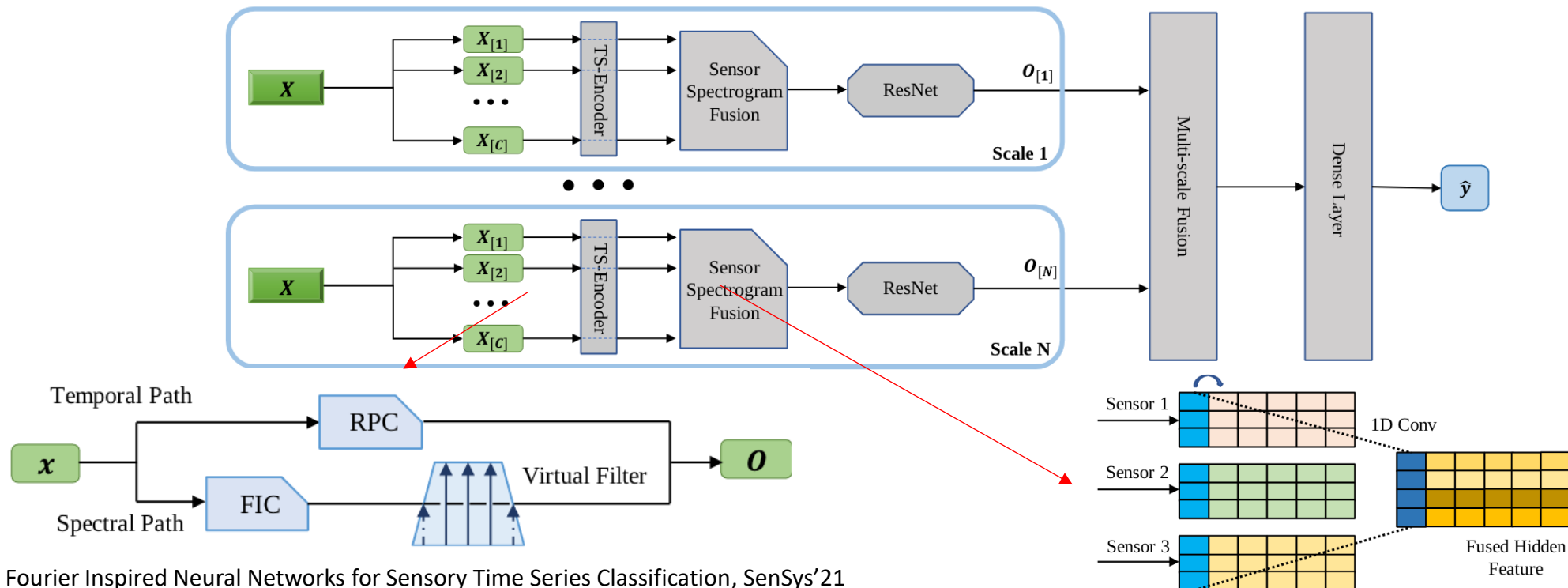
Wireless Data Representation

- Raw CSI is all you need?
 - Complex values
 - Amplitude only
 - Amplitude + phase
 - I/Q-components



Wireless Data Representation

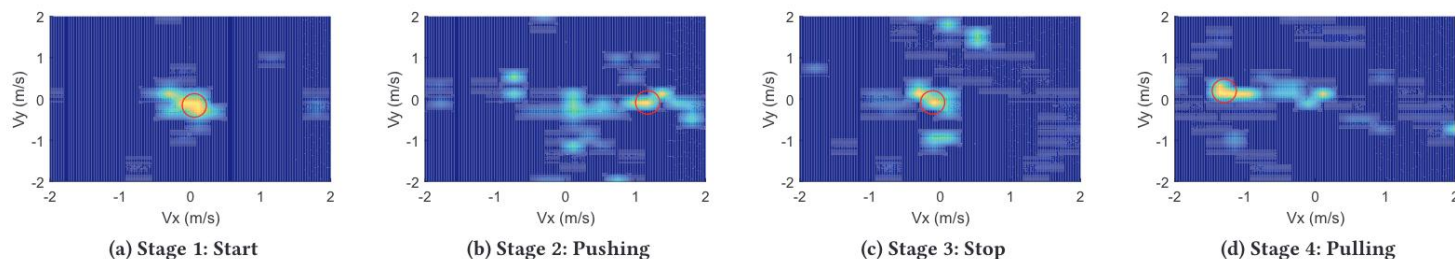
- DFS: Time-Frequency Spectrograms
 - integrate STFT into neural networks by initializing convolutional filter weights as the Fourier coefficients.



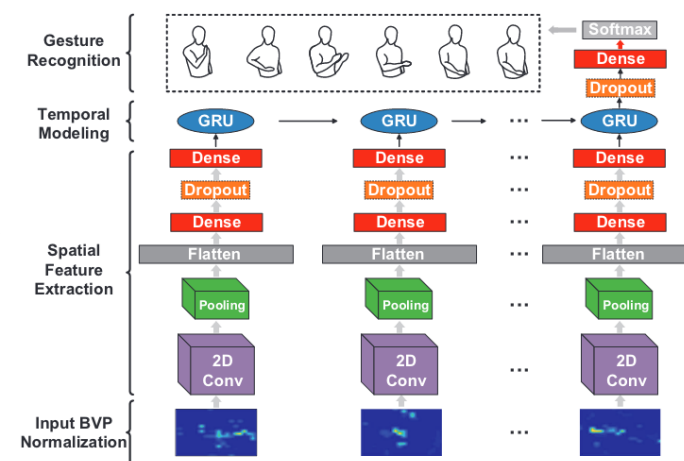
CSI Data Representation

- Widar3: Body-coordinate Velocity Profile (BVP)
 - Denoise via SP rather than completely relying on DL
 - Theoretically, domain-independent Body-coordinate Velocity Profile

For each activity,
obtain a BVP series:

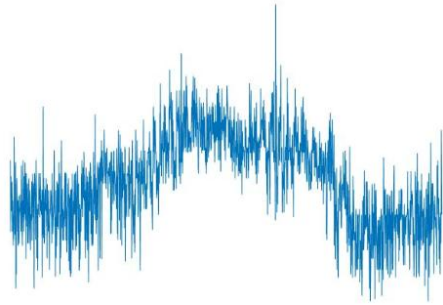


CNN+RNN for
classification

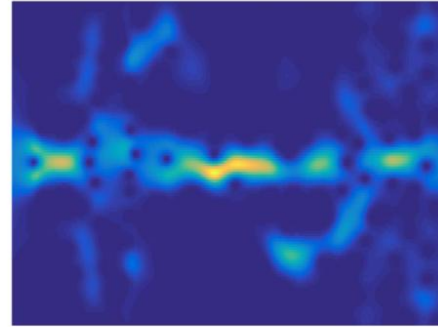


Widar3: BVP for Learning

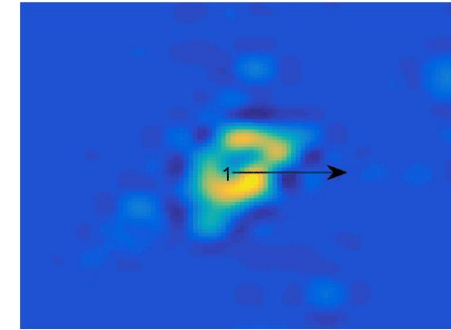
Domain-1
orientation #1
position #1
environment #1



CSI in Domain-1

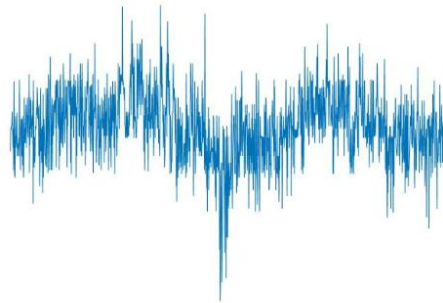


DFS in Domain-1

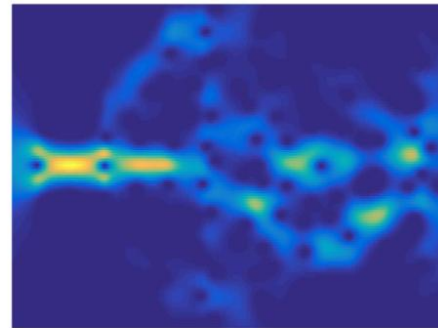


BVP in Domain-1

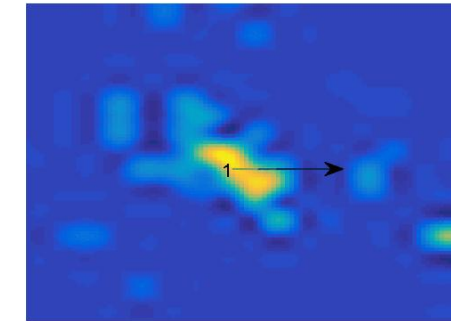
Domain-2
orientation #2
position #2
environment #2



CSI in Domain-2



DFS in Domain-2



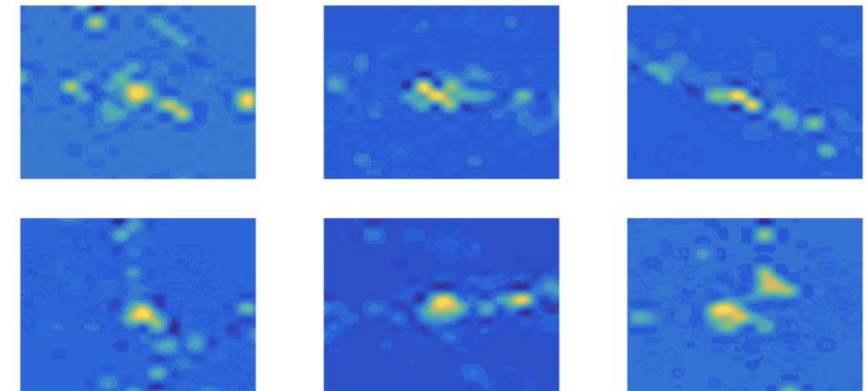
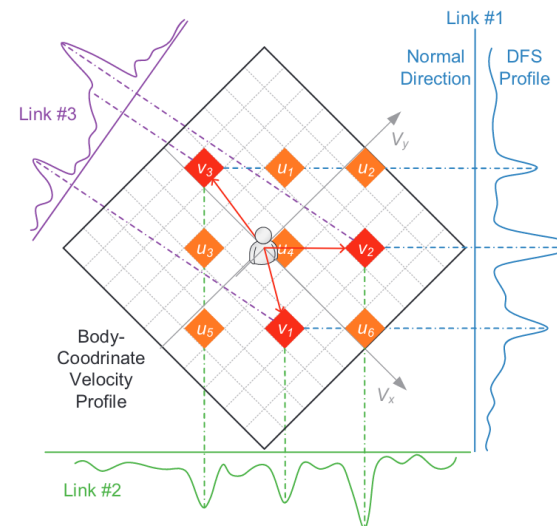
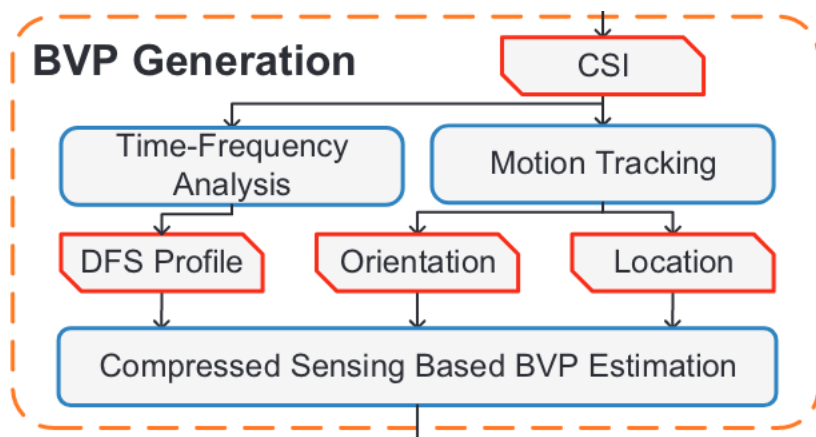
BVP in Domain-2



Wireless Data Representation

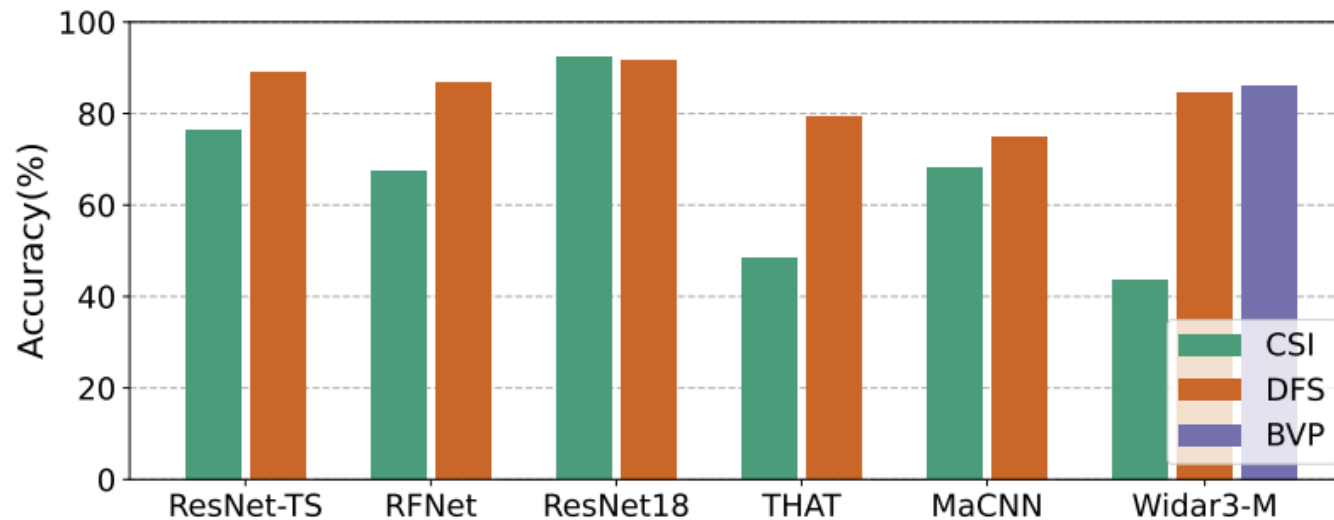
- BVP

- Theoretically, domain-independent Body-coordinate Velocity Profile
- Denoise via SP rather than completely relying on DL
- However, BVP relies on multiple links and imposes heavy computation



Wireless Data Representation

- What data representation should we use?
 - Still ad-hoc, no universal choice today
- Results of a preliminary comparison study



Performance comparison using CSI, DFS, and BVP as inputs on different models on Widar3 dataset.

Data Augmentation

- Data augmentation

- A set of techniques that artificially inflate the training samples from existing data.
- Increase the amount of data without collecting more
- Has been a common practice and proved effective in CV field

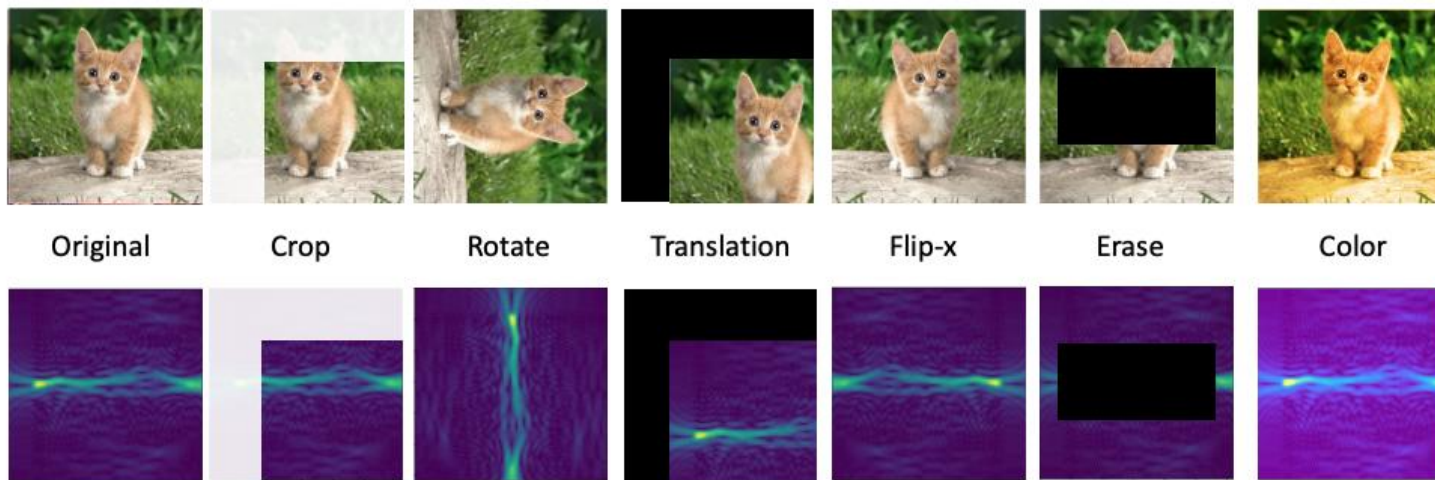
- Image data augmentation

- Random transformation: flipping, cropping, erasing, rotation, translation, color space transformation, neural style transfer, etc.



Data Augmentation

- Apply image data augmentation to radio data?
 - Radio data, even in the format of images, has different physical meanings
 - * rotating a spectrogram will reverse the time and frequency dimensions
 - * flipping it will reverse the time series (e.g., a pull gesture can become a push)
 - * random erasing can remove the critical part that has frequency responses
 - * changing the color space does not bring any new information

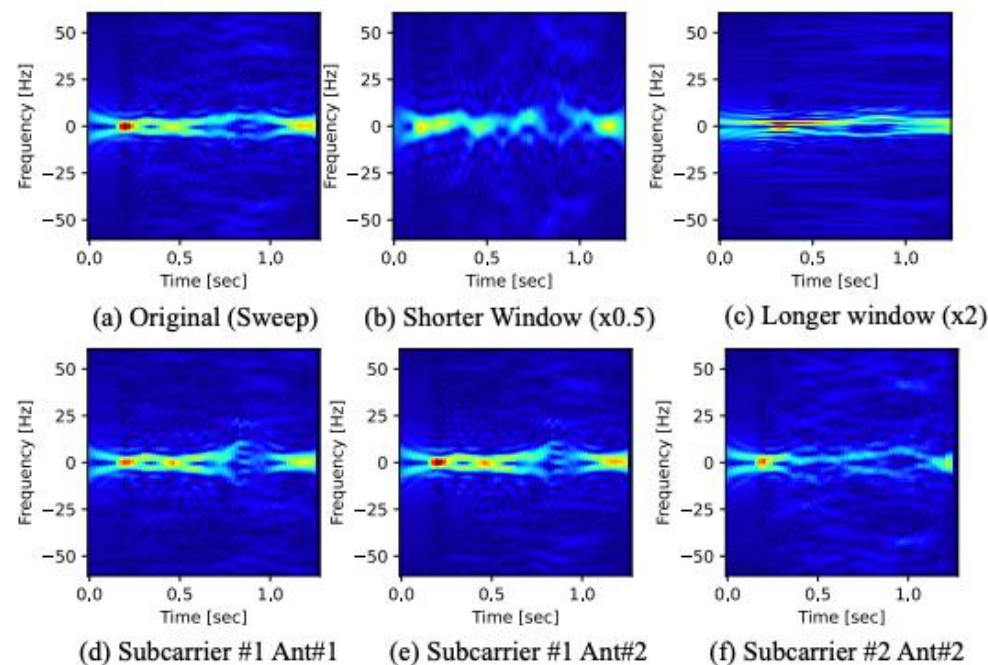
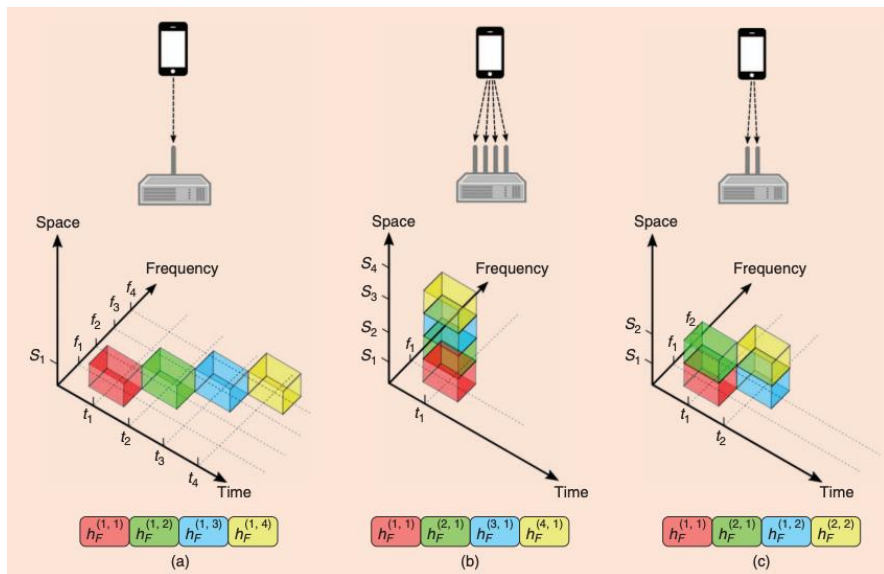


Make little sense, but
may still benefit accuracy
(given certain model and
available benchmarks)!

Data Augmentation

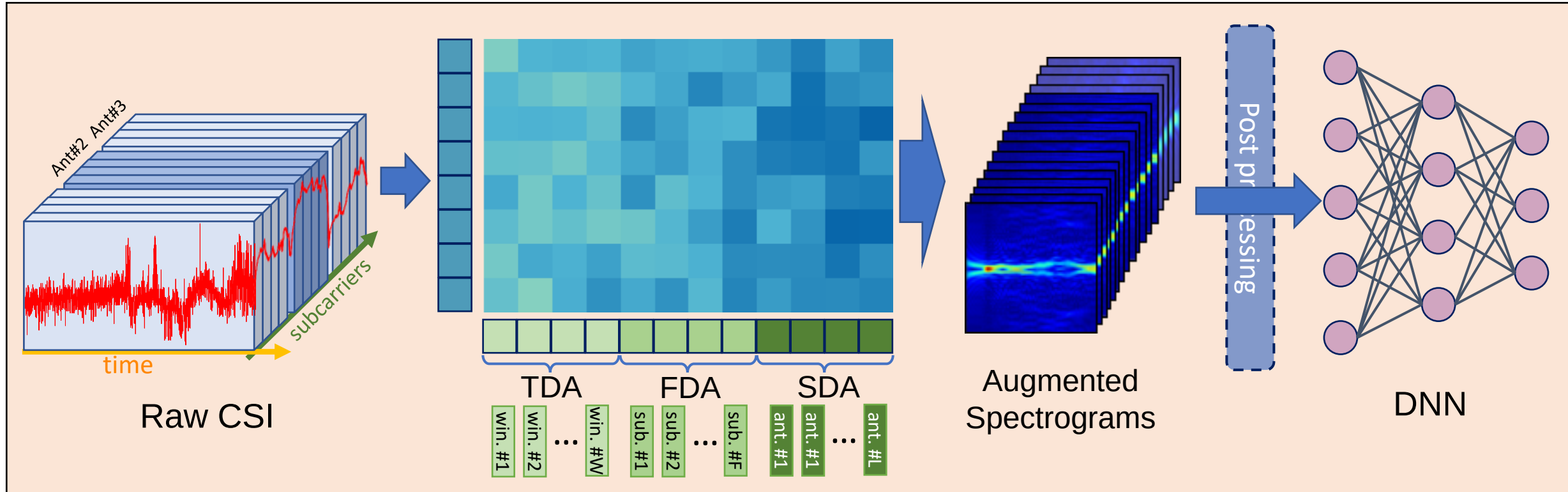
- Physical Data Augmentation

- Previously: PCA for dimension reduction → one single representative spectrograms per tx-rx link
- Recall diversity



Data Augmentation

- Physical Data Augmentation
 - Leverage data diversity to augment spectrograms!

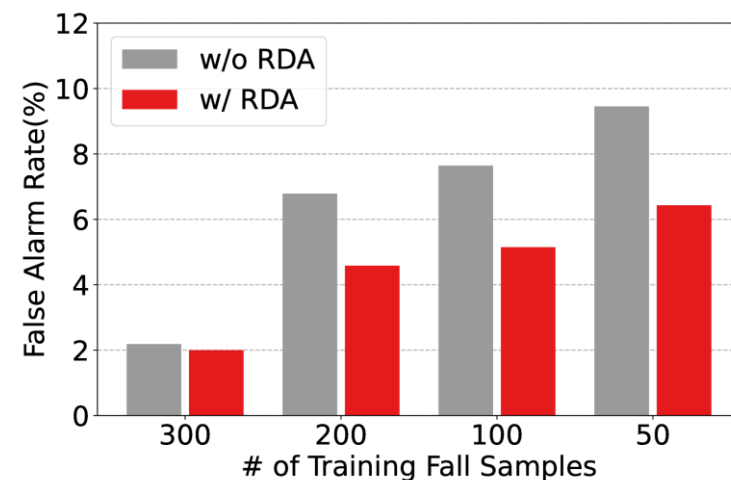
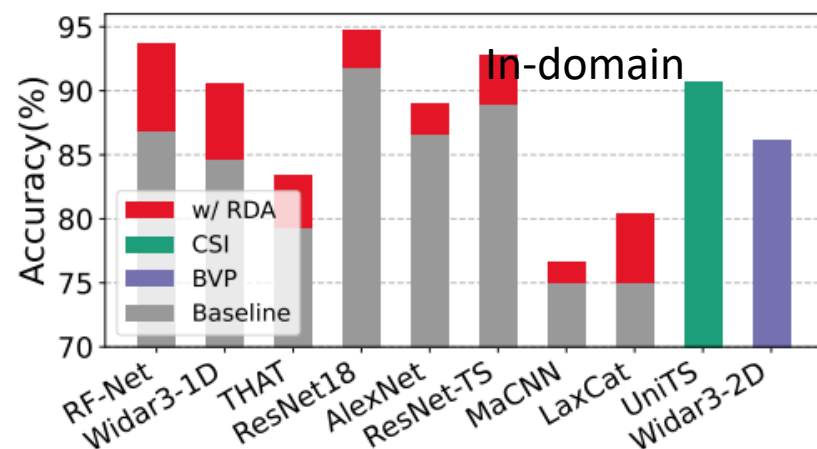
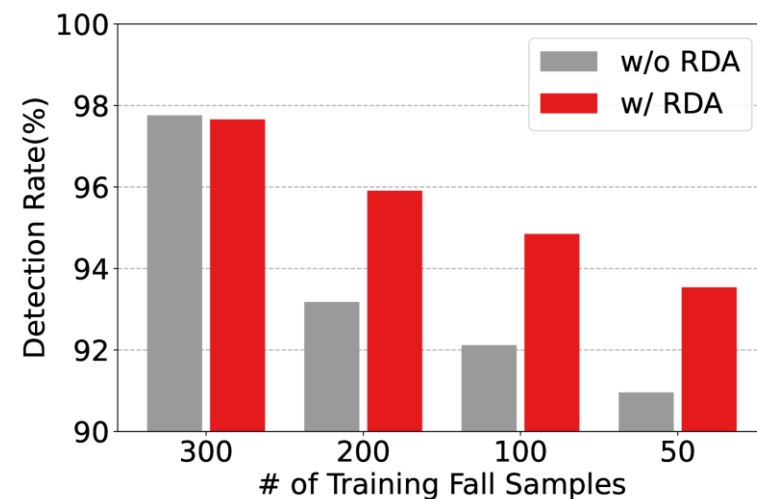
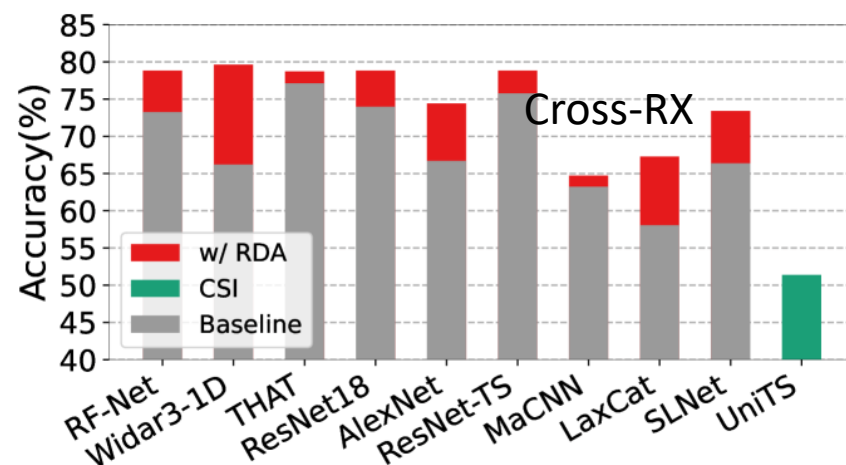


RFBoost: Physical Data Augmentation

- **Physical Data Augmentation**
 - **Time-domain Data Augmentation (TDA):** Using different windows for spectrogram generation
 - **Frequency/Space-domain Data Augmentation (FDA/SDA):** Combinations of different subcarriers/antennas
 - **Motion-aware Data Augmentation (MDA):** Motion-aware Random Erasing and Shifting
- All augmented samples are from physical observations



RFBoost: Performance



Gesture recognition

Fall detection

RFBoost: Performance

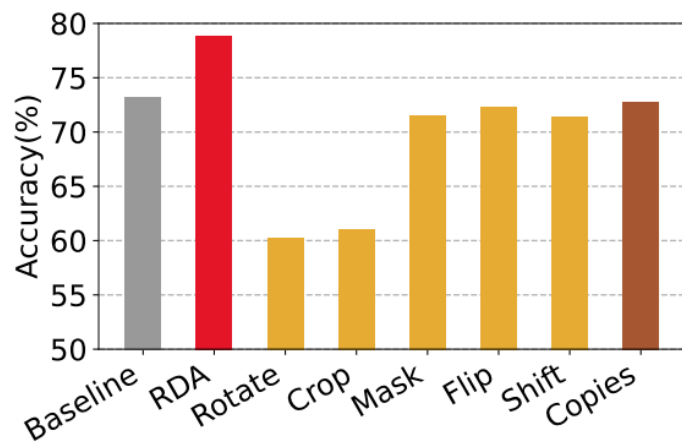


Fig. 18. Comparisons between IDA and RDA.

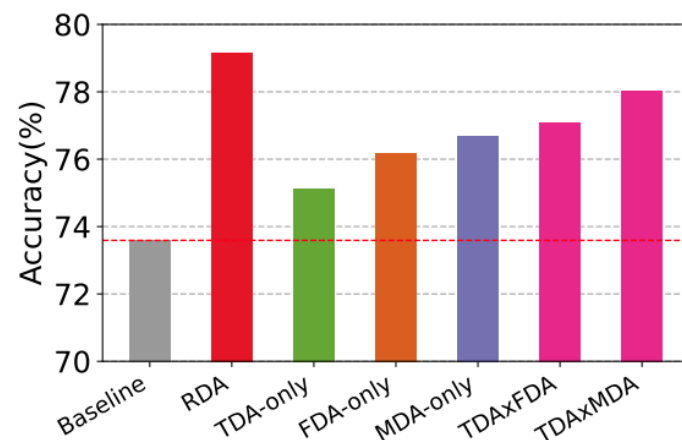
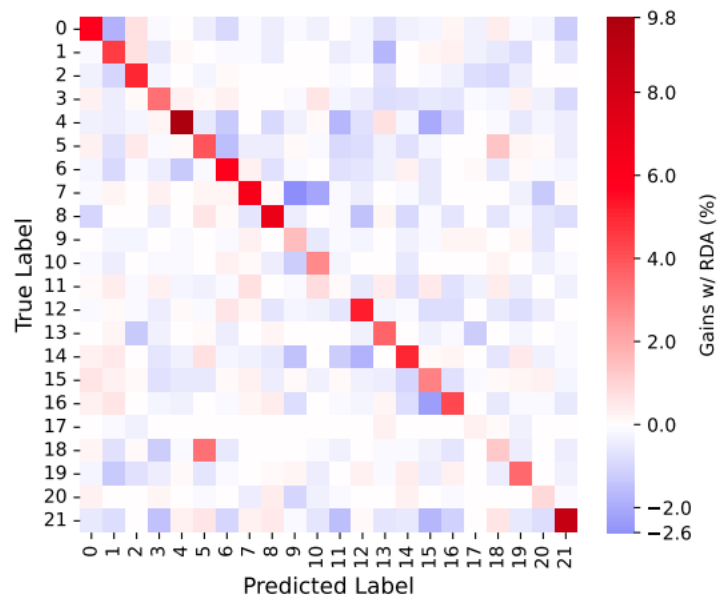
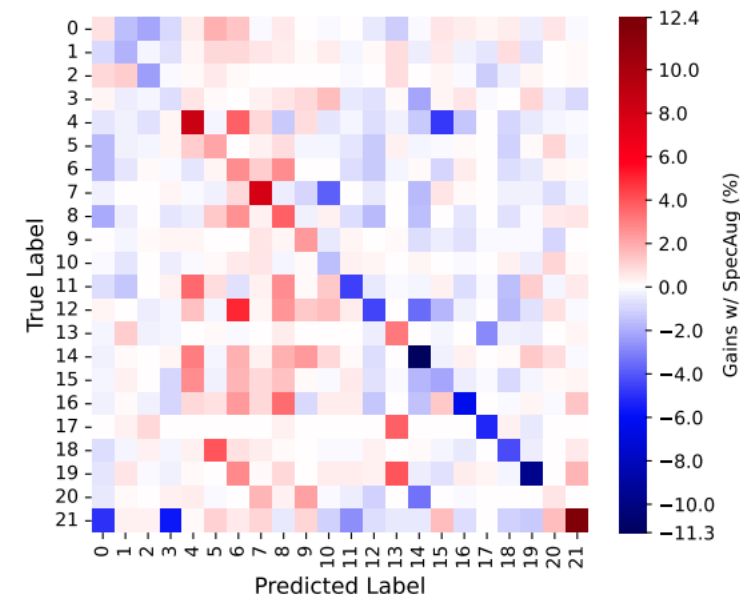


Fig. 19. Ablation study on different RDA policies.



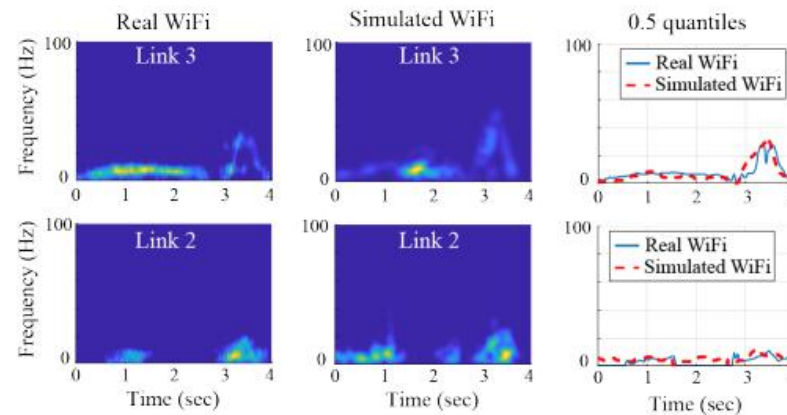
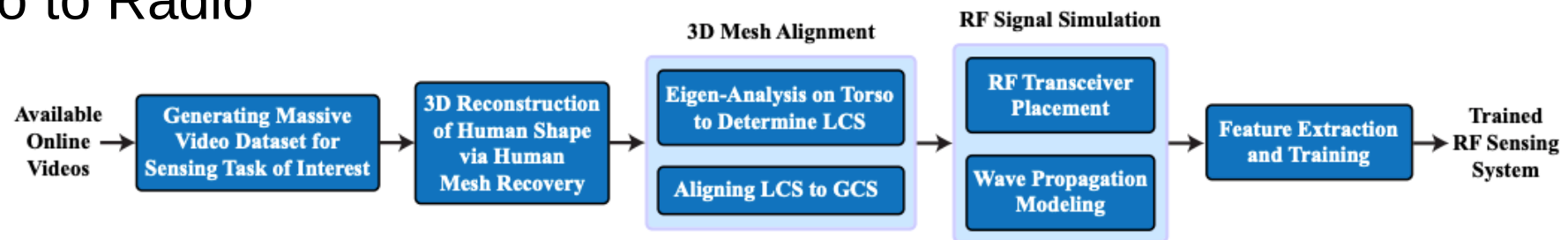
(a) Accuracy gains w/ RDA



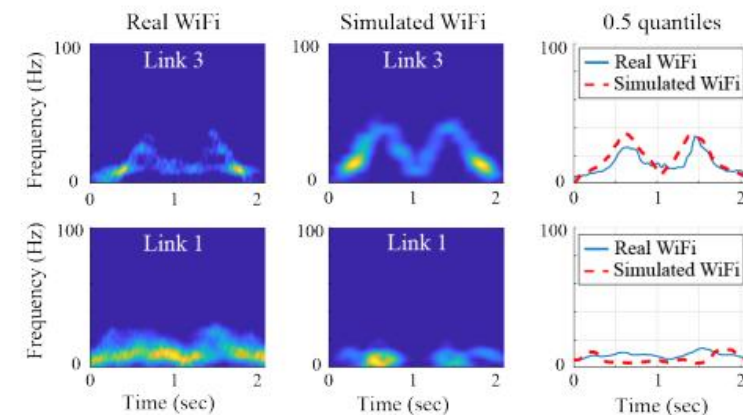
(b) Accuracy gains w/ SpecAugment

Data Synthesis/Generation

- Cross-modality training
 - Video to Radio



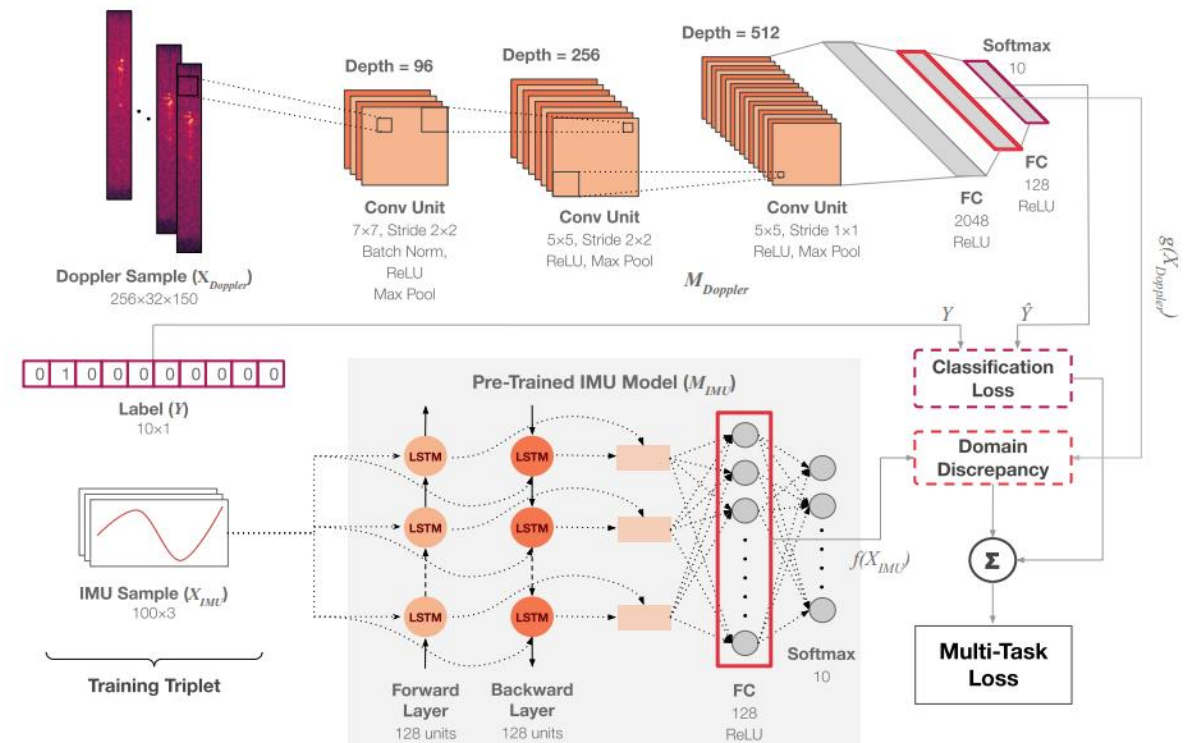
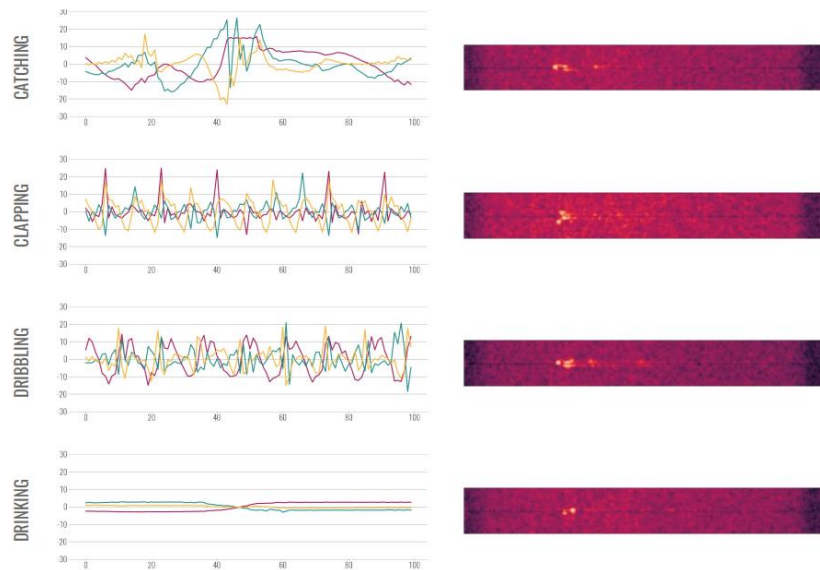
(a) Stiff-leg deadlift



(b) Forward Lunge

Data Synthesis/Generation

- Cross-modality training
 - IMU to Radio



RF-Diffusion: Radio Signal Generation via Time-Frequency Diffusion

- The AIGC era has arrived, encompassing various modalities.
 - Text: ChatGPT, LLaMA, ChatGLM, Claude, ...
 - Image: Stable Diffusion, DALL·E, Midjourney, ...
 - Video: Sora, Imagen Video, CogVideo, ...
 - Audio: Suno, AudioCraft, WaveNet, ...
- Generative AI has mastered most modalities, except the RF signal.

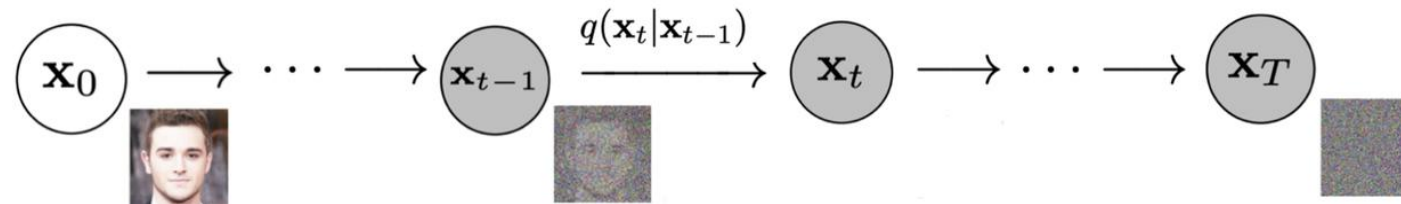
How about building an **RF Generative Model**?



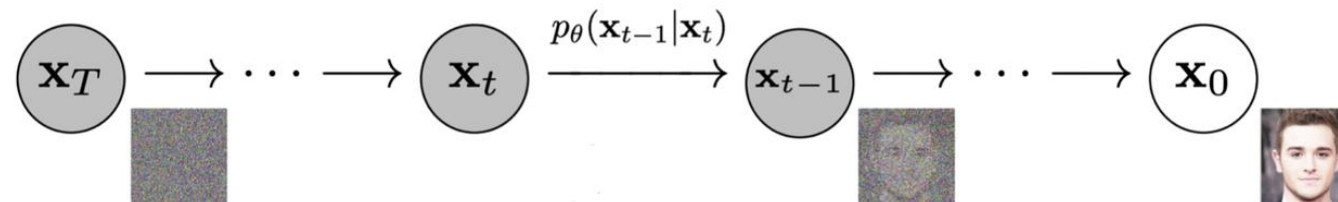
What is the diffusion process?

Traditional Diffusion Model performs the following two steps:

- **Forward Process:** gradually inject Gaussian noise to destruct the original data.



- **Reverse Process:** reconstruct the original distribution from noised data step by step.



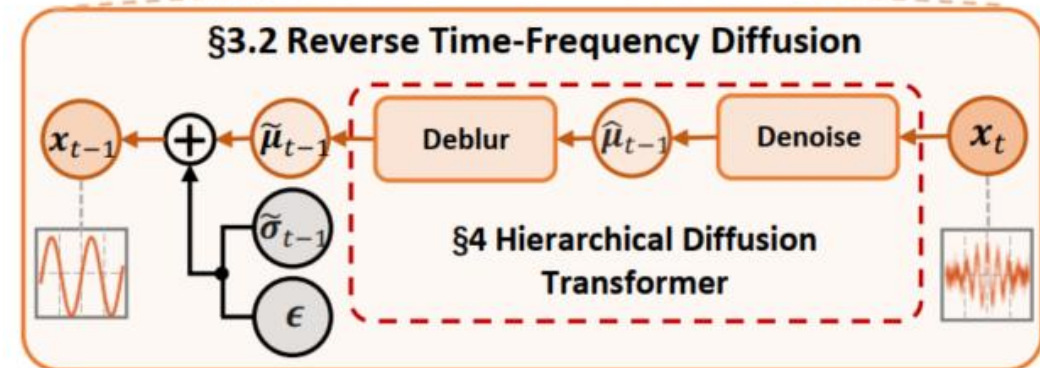
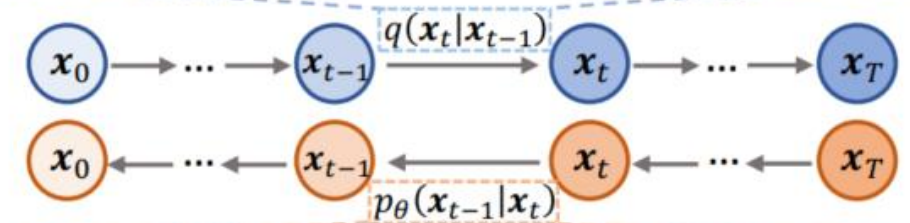
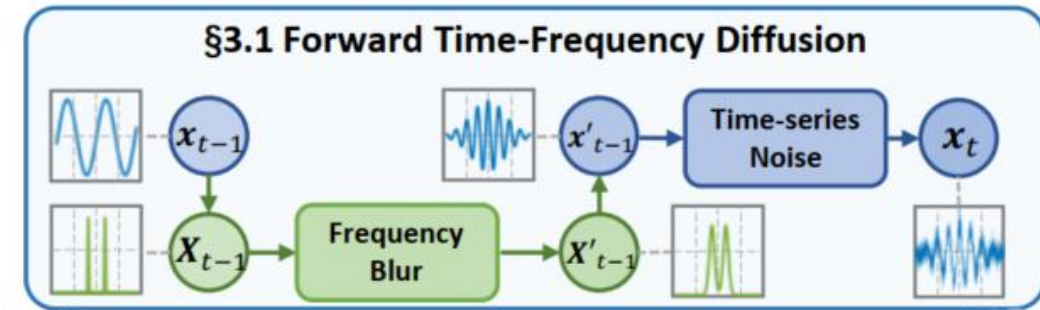
A well-trained **Diffusion Model** recover a learned data distribution from a random Gaussian sample, thereby achieving data generation.

How to adapt Diffusion to RF?

From the **theoretical** perspective, we propose the **Time-Frequency Diffusion**:

- **Forward Process:** iteratively add Gaussian noise in the time domain, while blurs the spectrum in the frequency domain.
- **Reverse Process:** reconstruct x_{t-1} by denoising x_t in the time domain, while deblurring in the frequency domain.

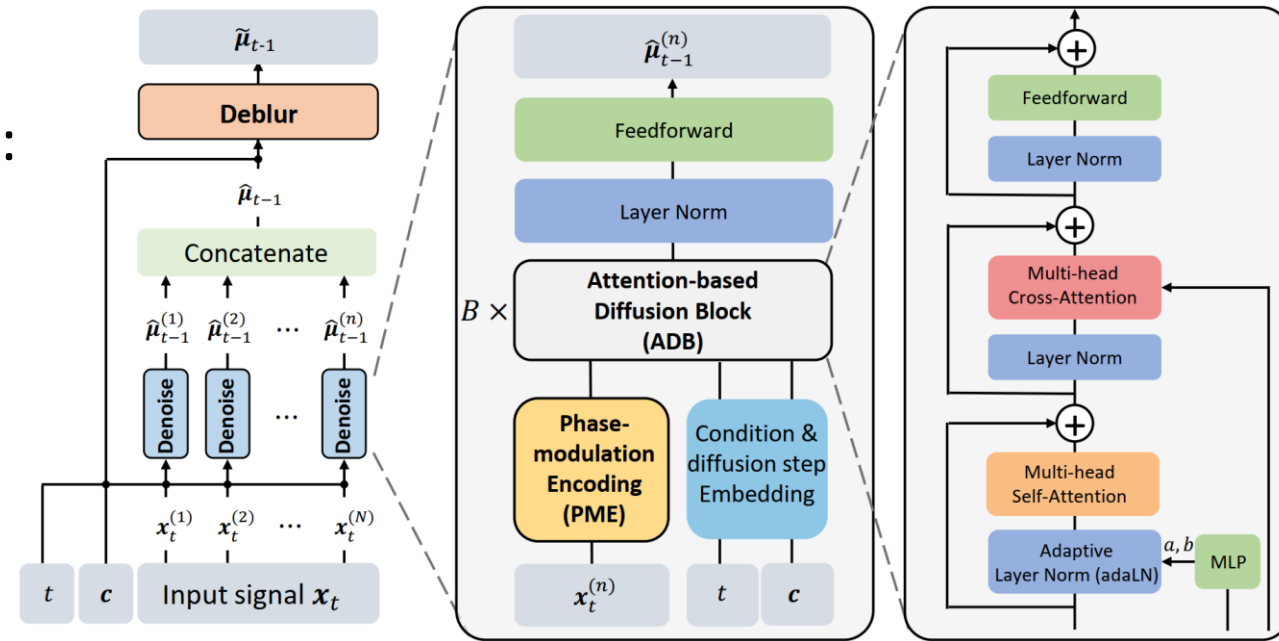
 Time-Frequency Diffusion emphasizes the **time-domain amplitude accuracy** and the **frequency-domain continuity**



How to adapt Diffusion to RF?

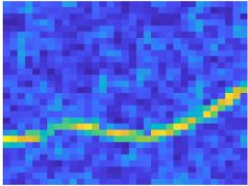
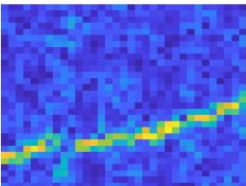
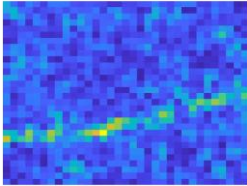
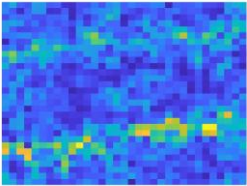
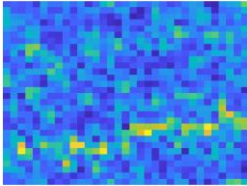
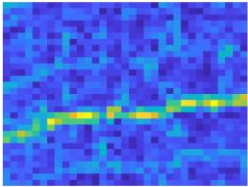
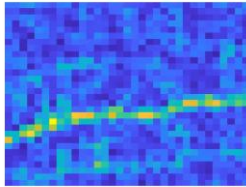
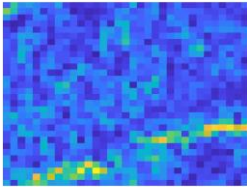
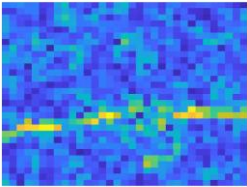
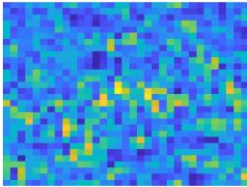
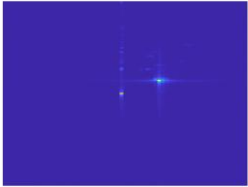
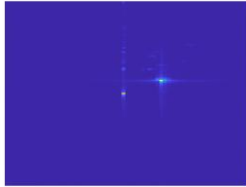
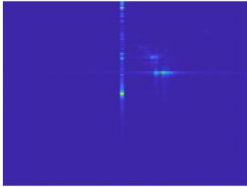
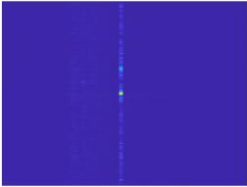
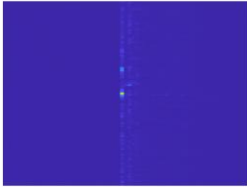
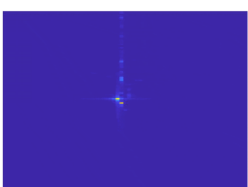
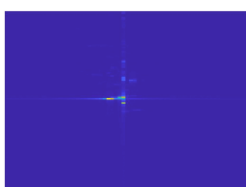
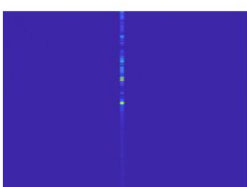
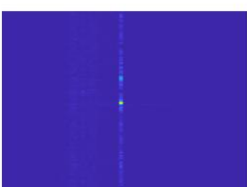
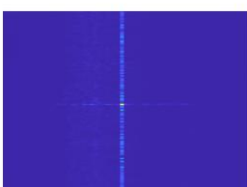
From the **implementation** perspective, we propose **Hierarchical Diffusion Transformer**:

- **Hierarchy**: HDT is divided into 2 stages, the denoising and deblurring stages.
- **Complex-valued design**: modify the attention module and feed-forward module to adapt to complex signals.
- **Phase modulation**: a positional encoding scheme tailed for complex-valued signals.



HDT adopts a **hierarchical architecture** to decouple noise and spectrum blur.

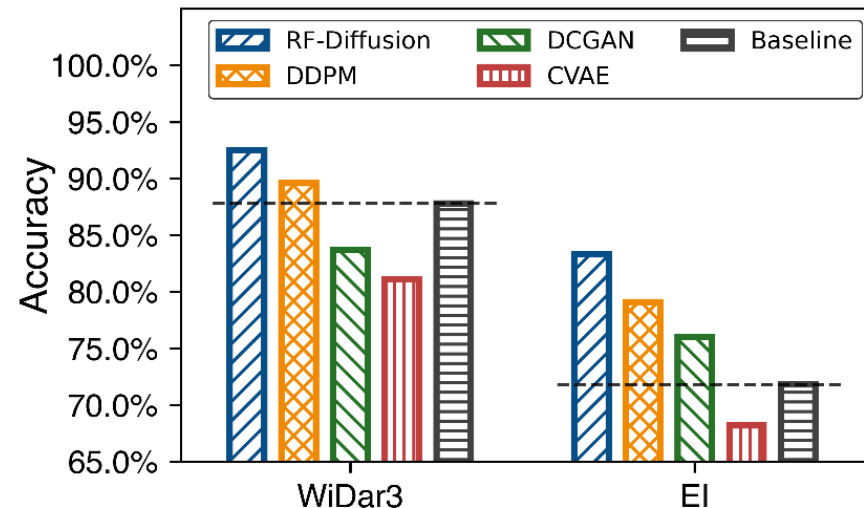
Evaluation Results

		Authentic	RF-Diffusion	DDPM	DCGAN	CVAE
Wi-Fi CSI (Doppler Frequency Spectrum)	Sample ①					
	Sample ②					
FMCW (Range- Velocity Doppler)	Sample ①					
	Sample ②					

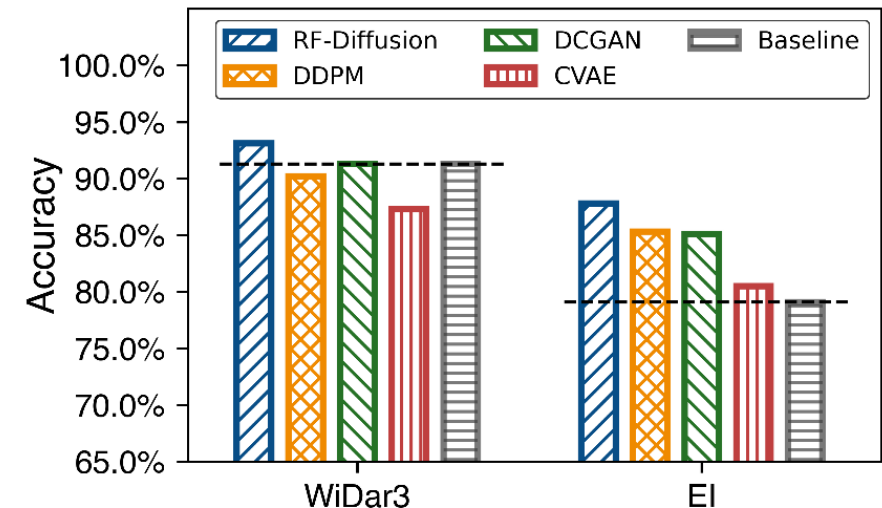
Case Study: Sensing data augmentation

Mixing synthesized wireless data with the original training set to jointly train gesture recognition models results in performance improvements:

- In cross-domain scenarios, Widar and EI saw increases of 4.7% and 11.5%, respectively;
- In in-domain situations, Widar and EI experienced gains of 1.8% and 7.5%, respectively;



Cross Domain

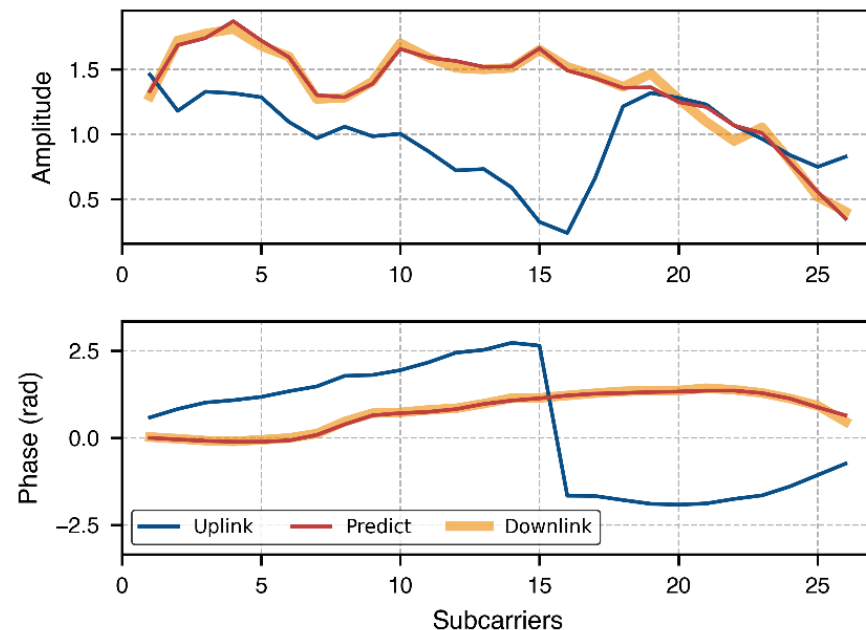


In Domain

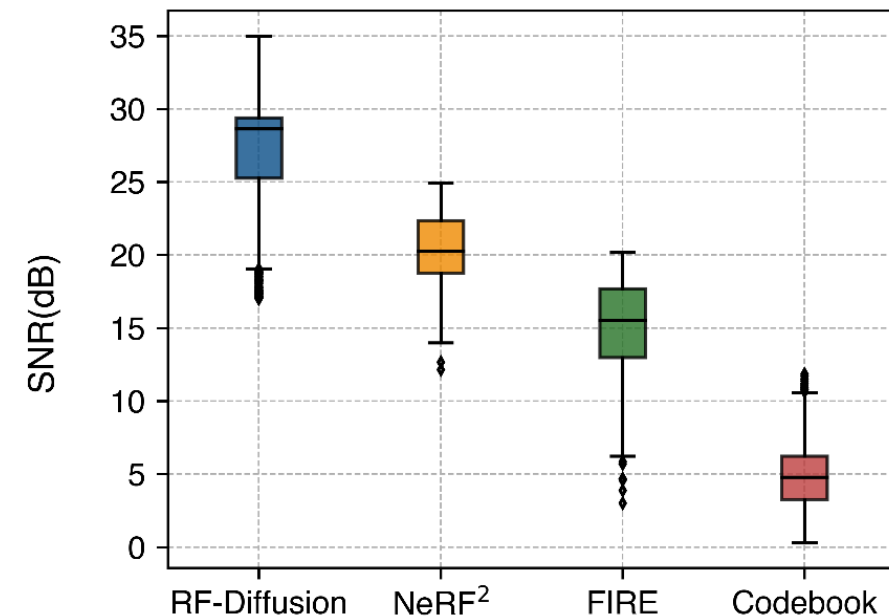
Case Study: Channel Prediction

Taking uplink channel CSI as a condition, RF-Diffusion can generate the downlink CSI:

- Compared to the SOTAs, RF-Diffusion achieved over 5.9 dB performance gain;
- The MSE of channel estimation has been reduced by about 70%;



Predicted CSI amplitude and phase



SNR Performance

Data Synthesis/Generation

- GenAI for wireless data generation is a promising direction
- NeRF (Neural Radiance Field) for Radio Frequency
- Open question:
 - How to generate large-scale data?
 - How to generate large-scale, high-quality data?

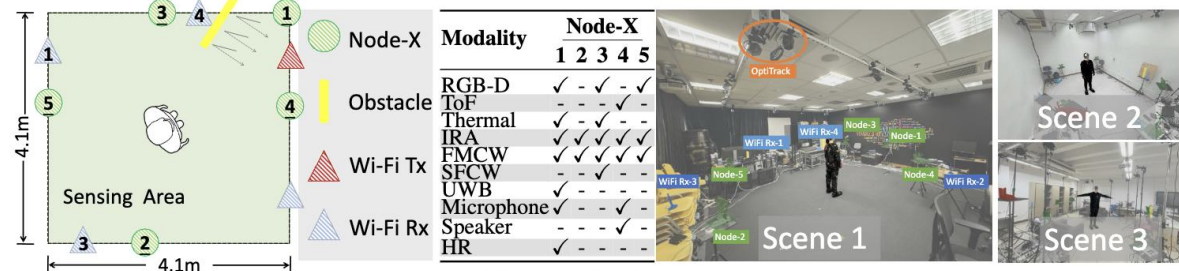


OctoNet

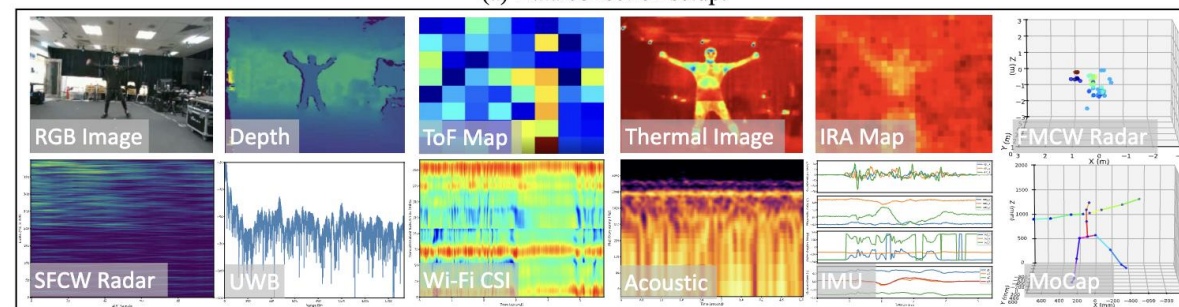
- Large-Scale Multi-Modal Dataset for Human Sensing
 - **13 heterogeneous modalities** (including RGB, depth, thermal cameras, infrared arrays, audio, millimeter-wave radars, WiFi, IMU)
 - **41 participants** under **multi-view** sensor setups
 - over **67.72M synchronized frames**.
 - **62 daily activities** spanning structured routines, freestyle behaviors, human-environment interaction, healthcare tasks
 - **all modalities are annotated by high-fidelity Mocap 3D pose labels**

<https://huggingface.co/datasets/hku-aiot/OctoNet>

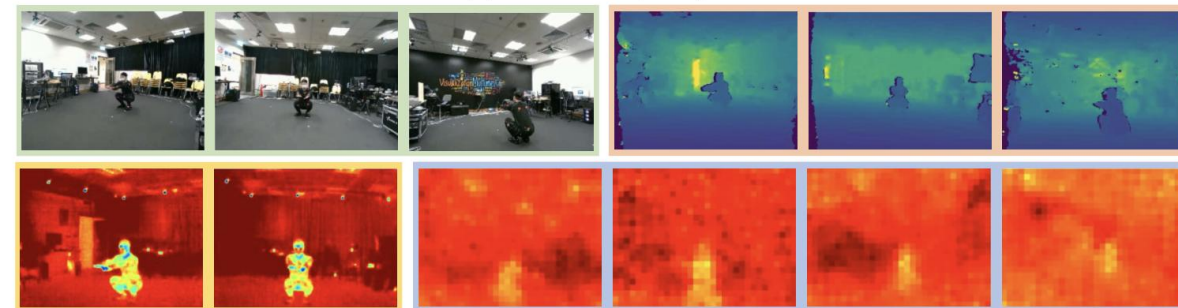
<https://github.com/aiot-lab/OctoNet/>



(a) Data collection setup.



(b) Multi-modality samples.



(c) Multi-view samples.

OctoNet

- A Large-Scale Multi-Modal Dataset for Human Activity Understanding
- Grounded in Motion-Captured 3D Pose Labels

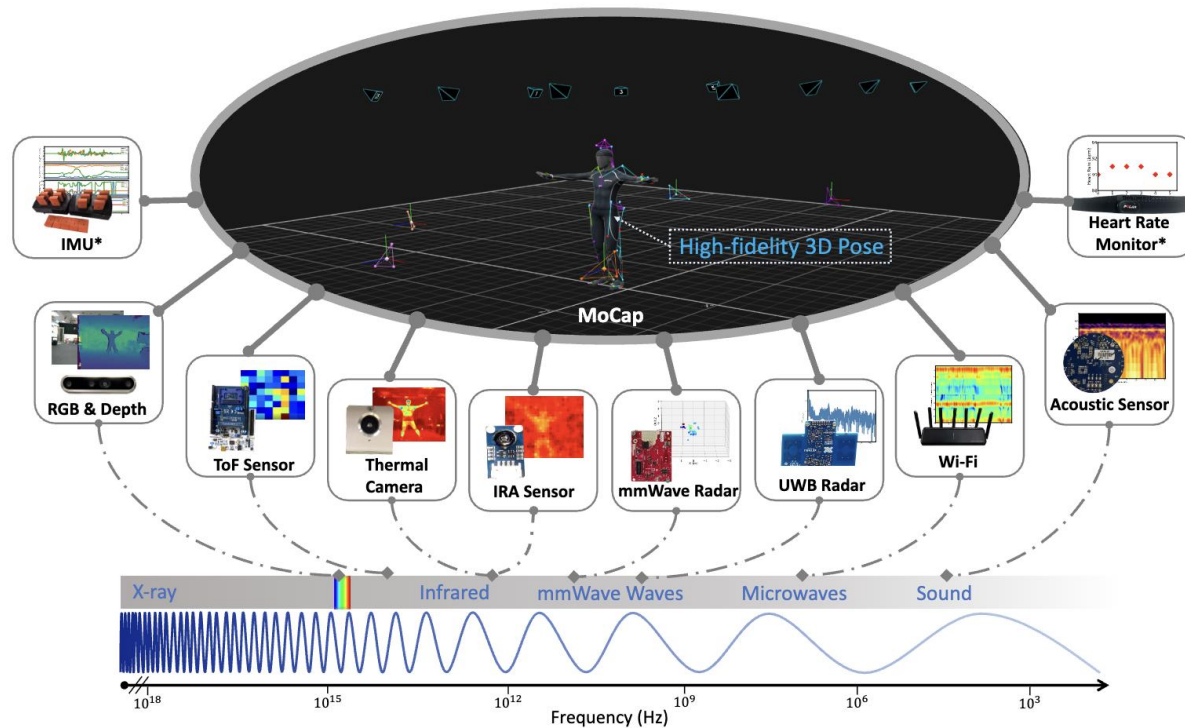


Table 1: Summary of existing single- and multi-modality datasets. OctoNet provides both action labels and high-fidelity 3D whole-body keypoint (3DKP) annotations.

Dataset	Modalities											Annotations		#Subj	#Act	#Seq	#Frame
	RGB-D	ToF	Thermal	IRA	mmWave	UWB	Wi-Fi	Acoustic	IMU	HR	MoCap	3DKP	Action				
CMU Panoptic [24]	✓	-	-	-	-	-	-	-	-	-	-	✓	-	8	5	65	154M
NTU RGB+D [54]	✓	-	-	-	-	-	-	-	-	-	-	✓	✓	40	60	56k	4M
Kinetics-700 [5]	✓*	-	-	-	-	-	-	-	-	-	-	-	✓	-	700	650.3k	-
KAIST-MP [18]	✓*	-	✓	-	-	-	-	-	-	-	-	-	-	1.18k	-	-	95.3k
PETS2017 [45]	✓*	-	✓	-	-	-	-	-	-	-	-	-	✓	-	10	36	-
CAMEL [13]	✓*	-	✓	-	-	-	-	-	-	-	-	-	-	-	-	26	44.5k
RF-Pose3D [79]	✓*	-	-	-	✓	-	-	-	-	-	-	✓	✓	>5	5	-	-
mmMesh [73]	✓*	-	-	-	✓	-	-	-	-	-	-	✓	✓	20	8	-	3k
mmBody [6]	✓	-	-	-	✓	-	-	-	-	-	✓	✓	✓	20	100	-	200k
Bocus UWB [3]	✓*	-	-	-	-	✓	-	-	-	-	-	-	✓	1	3	-	2.9M
Widar 3.0 [80]	-	-	-	-	-	-	✓	-	-	-	-	-	✓	17	22	-	17.8k
WiPose [23]	✓*	-	-	-	-	-	✓	-	-	-	-	✓	✓	10	16	-	96k
GoPose [51]	✓*	-	-	-	-	-	✓	-	-	-	-	✓	✓	10	>9	-	676.2k
UbiCoustics [26]	-	-	-	-	-	-	-	✓	-	-	-	-	✓	12	30	-	-
SAMoSA [38]	-	-	-	-	-	-	-	✓	✓	-	-	-	✓	20	26	1560	-
UTD-MHAD [7]	✓	-	-	-	-	-	-	-	✓	-	-	✓	✓	8	27	861	-
USC-HAD [77]	-	-	-	-	-	-	-	-	✓	-	-	-	✓	14	12	840	-
Total Capture [62]	✓*	-	-	-	-	-	-	-	✓	-	✓	✓	✓	5	4	60	1.9M
Stanford-ECM [39]	✓*	-	-	-	-	-	-	-	✓	✓	-	-	✓	10	24	113	-
Opportunity++ [8]	✓*	-	-	-	-	✓	-	-	✓	-	-	-	✓	4	43	24	-
mRI [2]	✓	-	-	-	✓	-	-	-	✓	-	-	✓	✓	20	12	300	160k
OPERAmet [4]	✓	-	-	-	-	✓	✓	-	-	-	-	✓	✓	6	6	61	-
MM-Fi [74]	✓	-	-	-	✓	-	✓	-	-	-	-	✓	✓	40	27	1080	320.8k
XRF55 [66]	✓	-	-	-	✓	-	✓	-	-	-	-	-	✓	39	55	42.9k	-
OctoNet (Ours)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	41	62	8.76k	67.72M

* Indicates datasets with RGB-only imagery (no depth sensor).

<https://huggingface.co/datasets/hku-aiot/OctoNet>
<https://github.com/aiot-lab/OctoNet/>

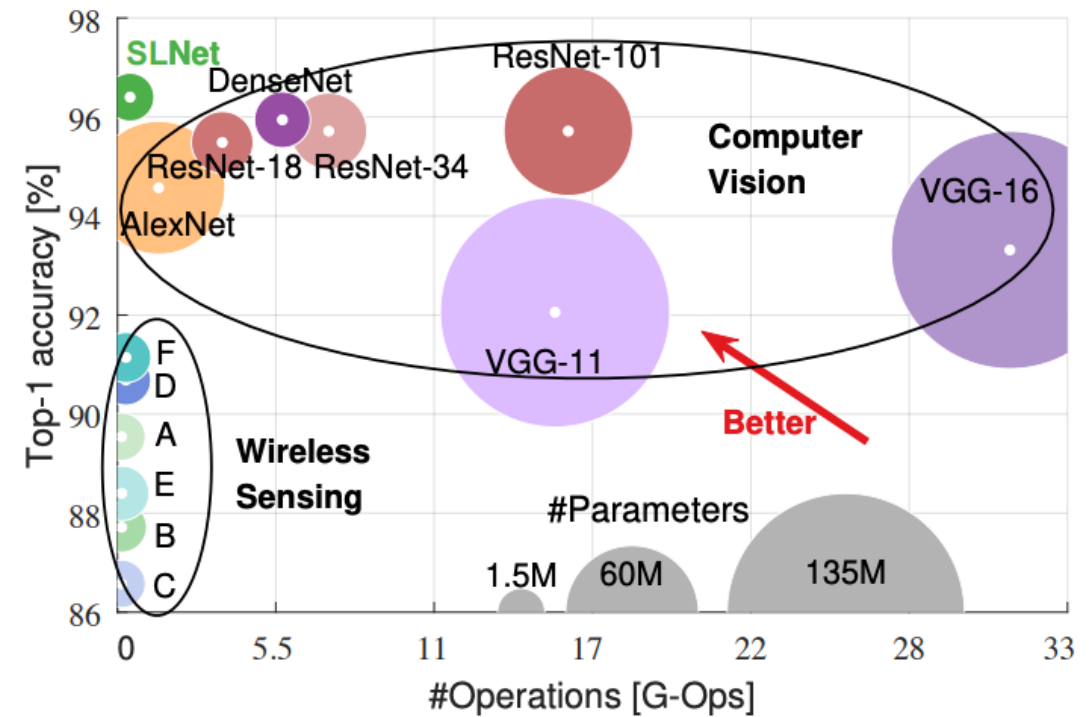
"OctoNet: A Large-Scale Multi-Modal Dataset for Human Activity Understanding Grounded in Motion-Captured 3D Pose Labels", NeurIPS'25

Chenshu Wu | HKU

SLNet: Spectrogram Learning Neural Network

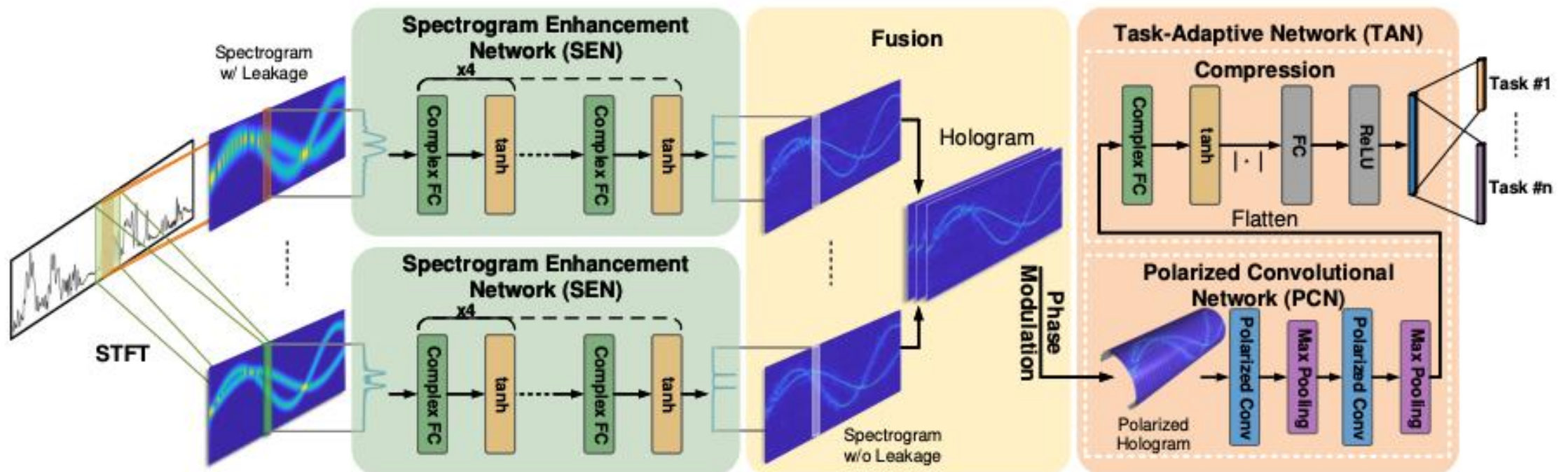
- Do we need separate design for DWS or not? And how?

- Many are based on mature CV models, accurate but quite big
- Existing DWS models are relatively small but less accurate



SLNet

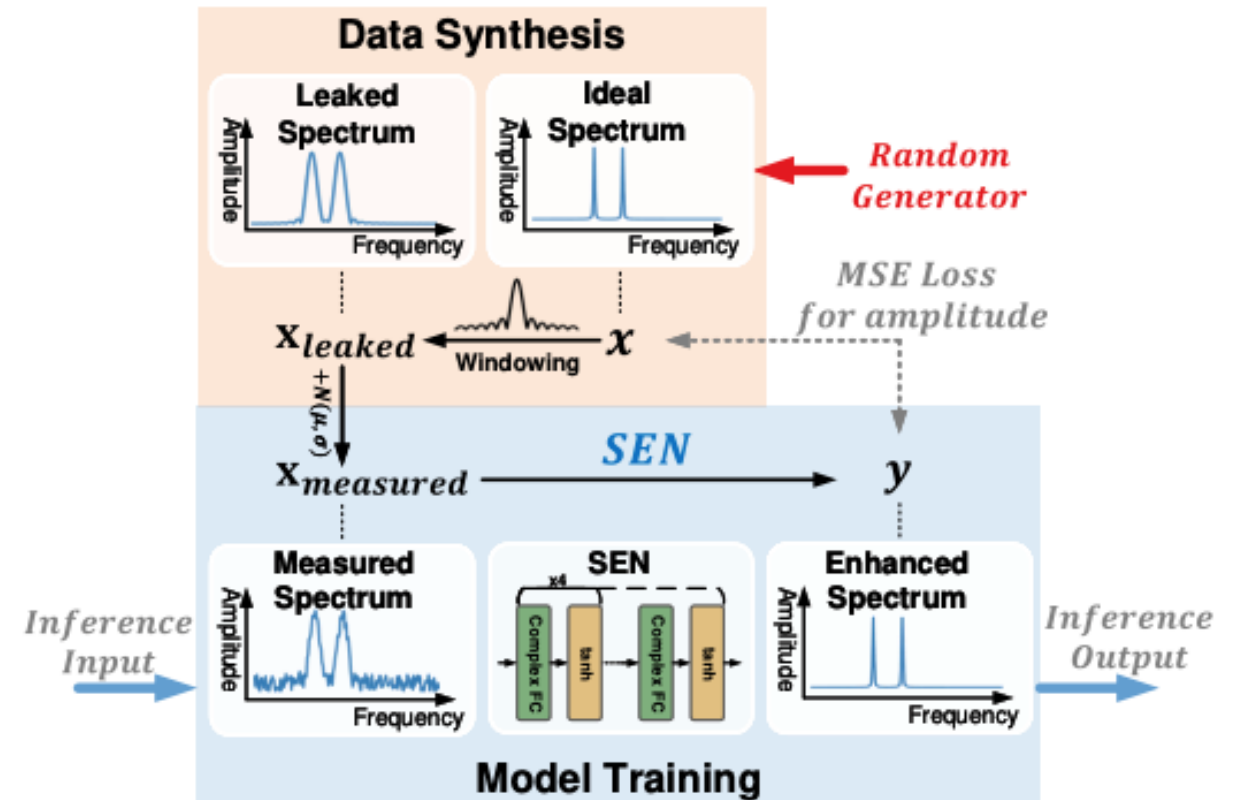
- An attempt for DWS models
 - Based on spectrogram learning



SLNet

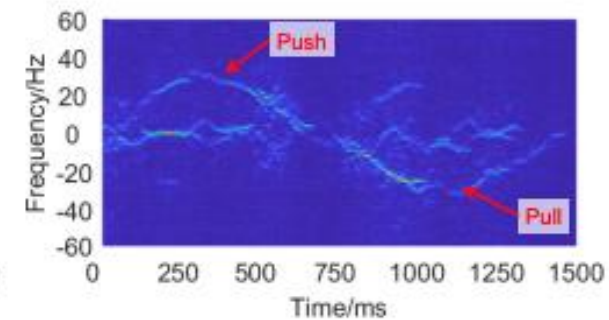
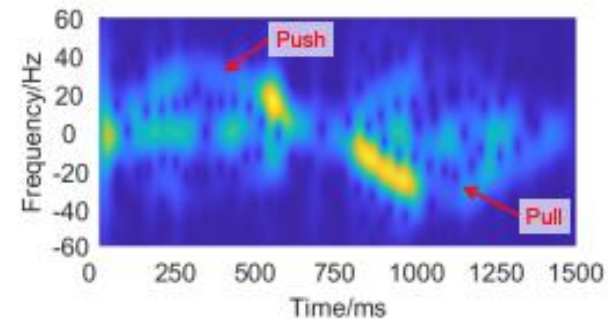
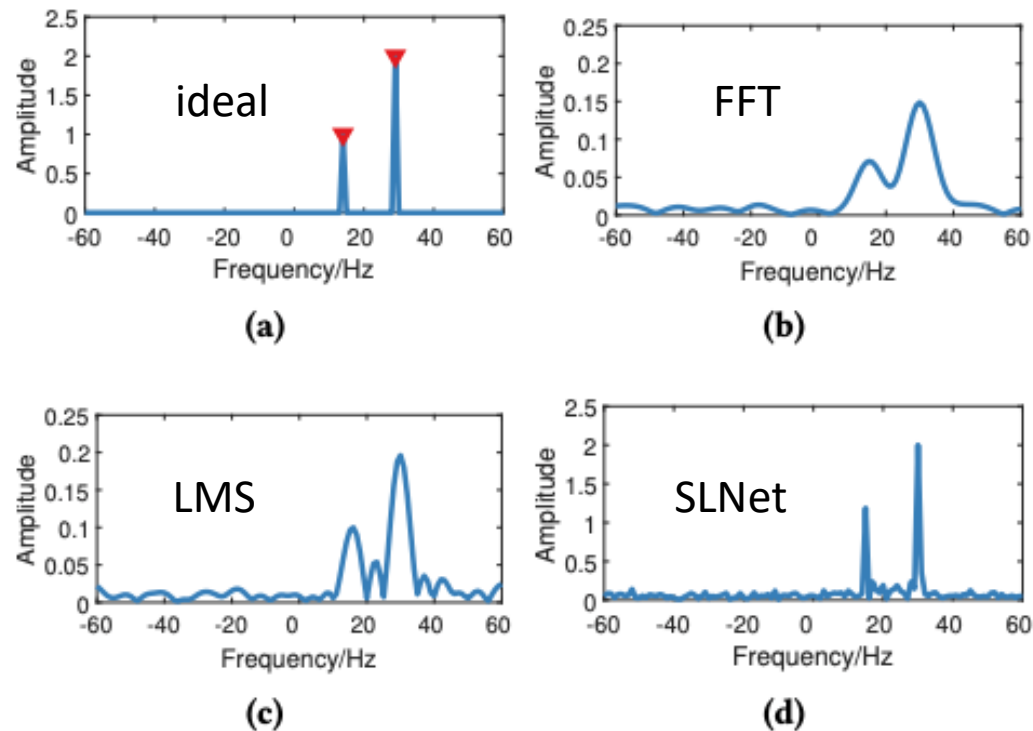
- How to generate high-fidelity spectrograms?

- Spectrograms generated by STFT suffer from spectral leakage, an inherent issue of FFT.
- SEN: Spectrogram Enhancement Network to learn the best function to minimize or nearly eliminate the leakage.



SLNet

- How to generate high-fidelity spectrograms?



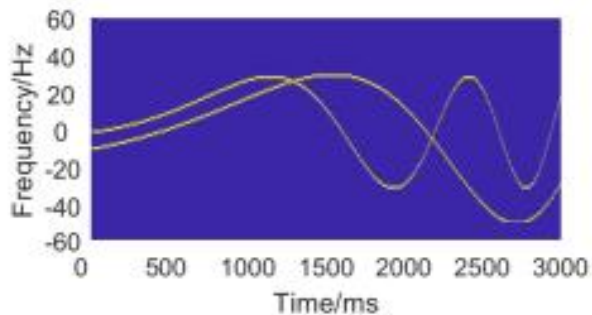
STFT spectrogram

SEN spectrogram

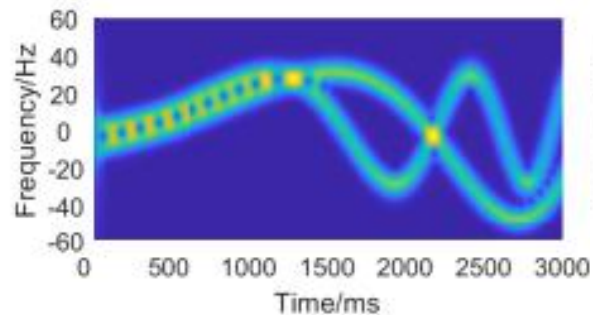
The spectrogram of a pushing and pulling gesture.

SLNet

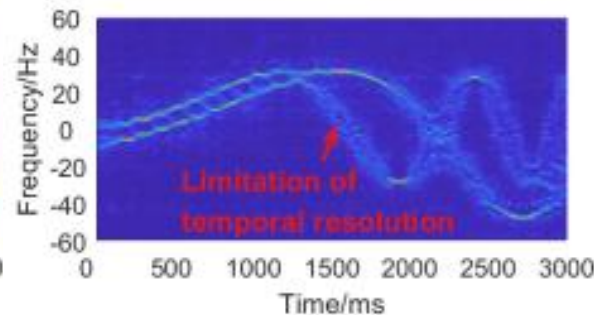
- How to trade-off time-frequency resolution?
 - Achieve high frequency resolution by using long windows
 - Capture fast-changing frequencies by using short windows
- Use a bank of sliding windows with different lengths
 - Concatenated as multiple channels, forming a multi-channel “hologram”



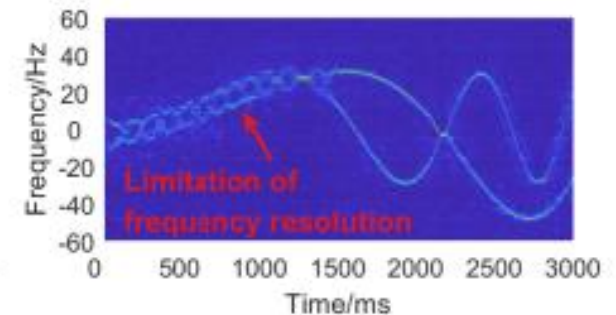
Ideal spectrogram



STFT spectrogram ($w=251$ ms)



SEN spectrogram ($w=251$ ms)



SEN spectrogram ($w=125$ ms)

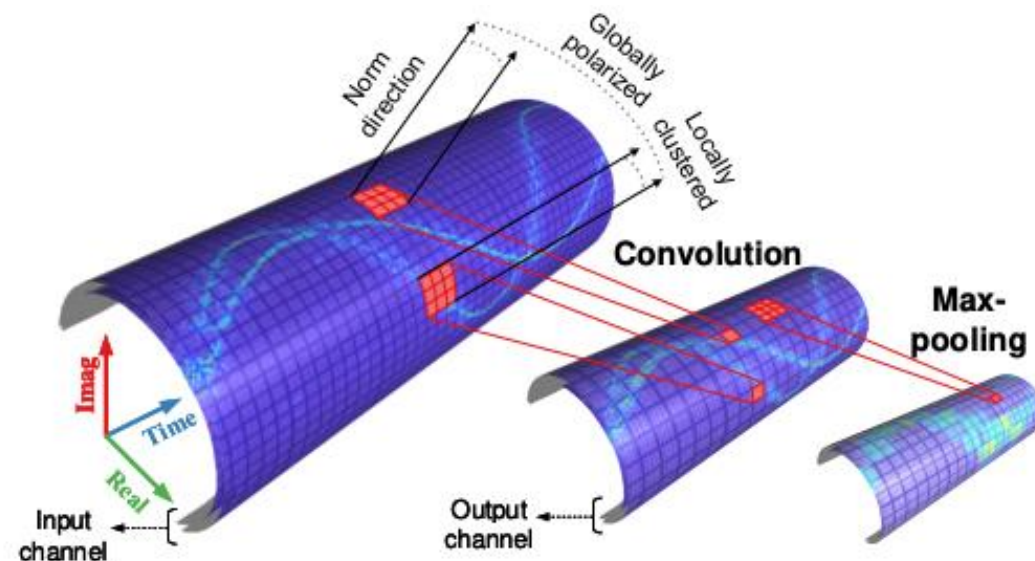
SLNet

- How does a model simultaneously preserve local dependency and global discrimination?
- Problem
 - CNN mainly learns local features irrespective of global locations of objects in an image
 - Not ideal for spectrogram learning, as the global locations, i.e., frequencies, are correlated with the physical properties of a person's activities, which is not shift-invariant.



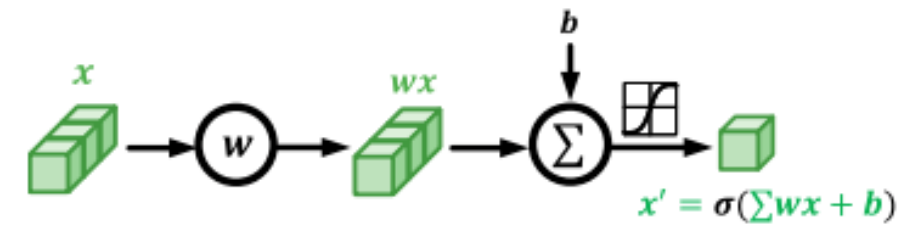
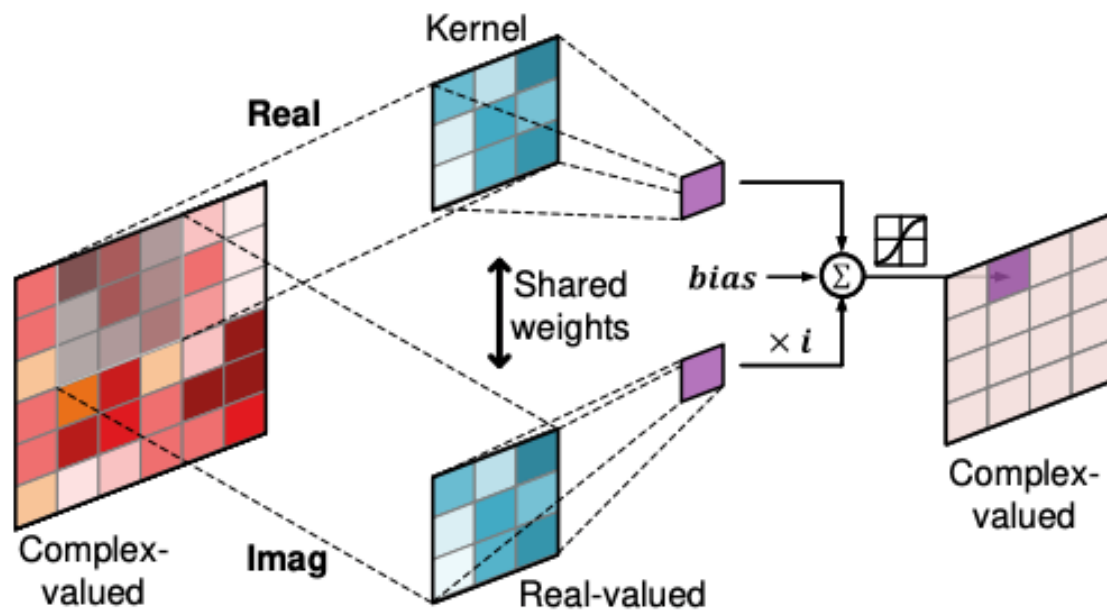
SLNet: Spectrogram Learning NN

- How does a model simultaneously preserve local dependency and global discrimination?
- **Solution: Polarized Convolutional Network**
 - Polarize the spectrograms via linearly modulated phase information
 - Adjacent frequency components have similar phases while the distant ones have discriminative phases
 - Use phases that vary linearly along the frequency dimension, making them locally unaltered while globally differentiated

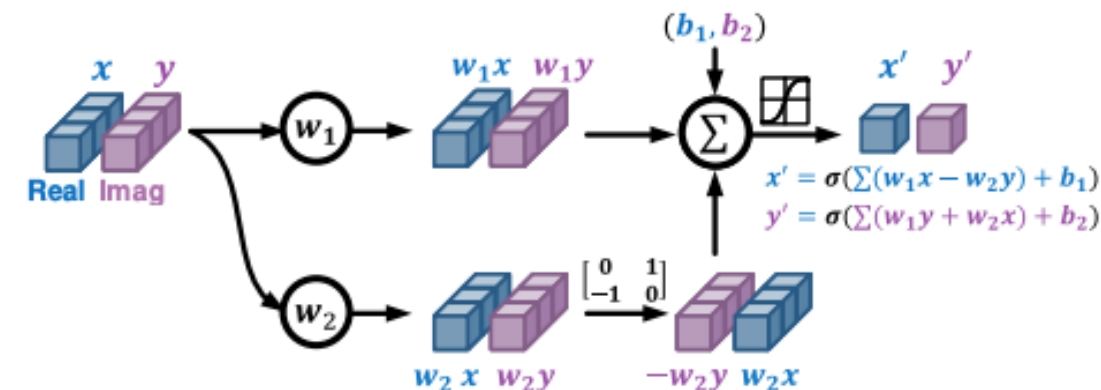


SLNet

- How to learn from the polarized complex-valued spectrograms?



real-valued vs. complex-valued neurons



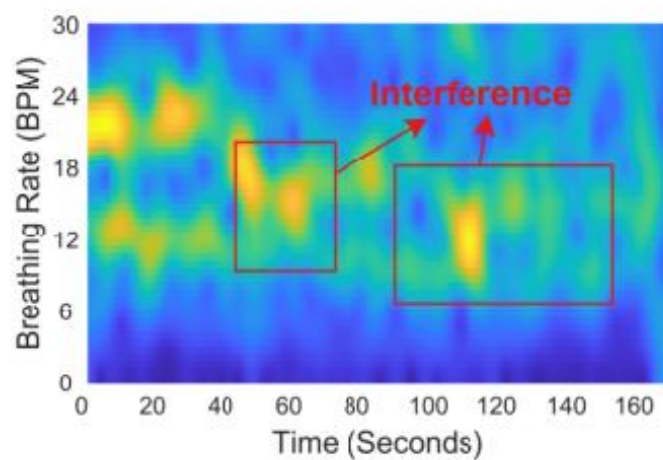
SLNet

- How well does it work?

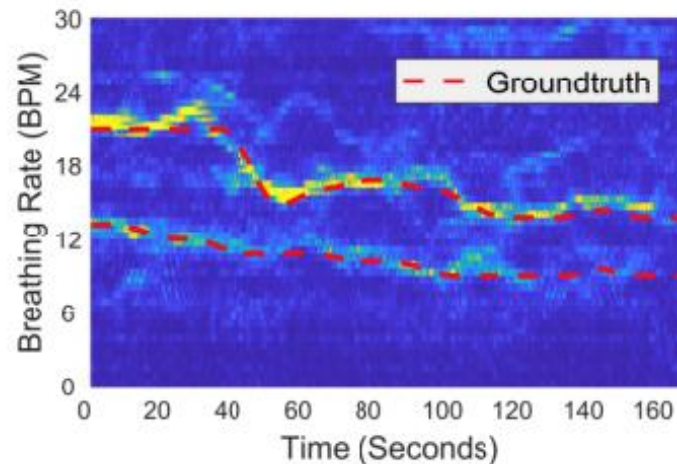
Modality	Ref.	Gesture	Gait	Fall ¹	Para ²
WiFi	[23, 90]	90.6%	95.1%	92.8%, 96.3%	1.07M
	[8, 22]	89.0%	96.6%	96.4%, 84.3%	2.72M
	[39, 79]	84.3%	83.3%	96.8%, 93.8%	5.77M
	[73] ³	78.9%	70.9%	95.5%, 96.8%	0.06M
FMCW	[87]	88.0%	95.4%	96.0%, 96.0%	1.06M
	[84, 86]	91.6%	96.4%	99.7%, 95.7%	2.76M
Acoustic	[30]	89.6%	95.4%	90.6%, 98.3%	6.08M
Vision	[40]	88.3%	90.1%	95.3%, 95.3%	128.8M
	[15]	91.9%	96.6%	97.0%, 95.6%	11.18M
	[20]	91.0%	97.7%	99.8%, 96.3%	6.96M
CVNN	[17, 32]	72.3%	96.0%	95.2%, 93.7%	115.6M
	[46]	92.0%	96.3%	98.4%, 93.8%	2.94M
WiFi	SLNet	96.6%	98.9%	99.8%, 97.2%	1.48M

SLNet

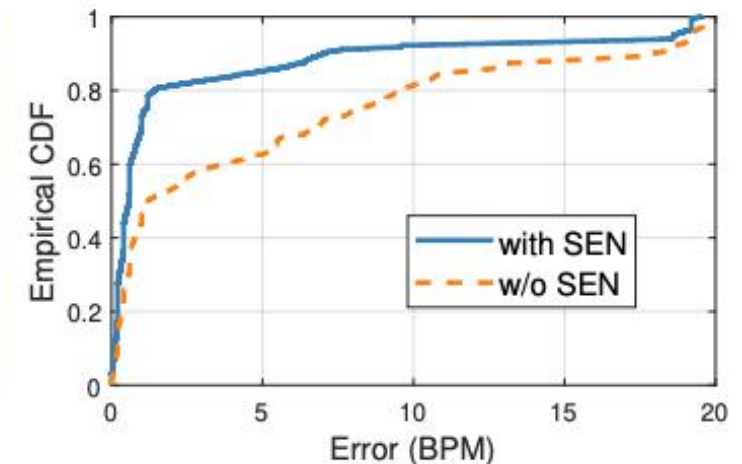
- How well does it work?
 - Multi-user breath estimation



(a) The raw spectrogram from traditional FFT. Spectral leakage causes severe interference for the close frequency components, making it hard to detect the breath rates of different people.



(b) The enhanced spectrogram with SEN. Frequency components can be clearly discriminated.

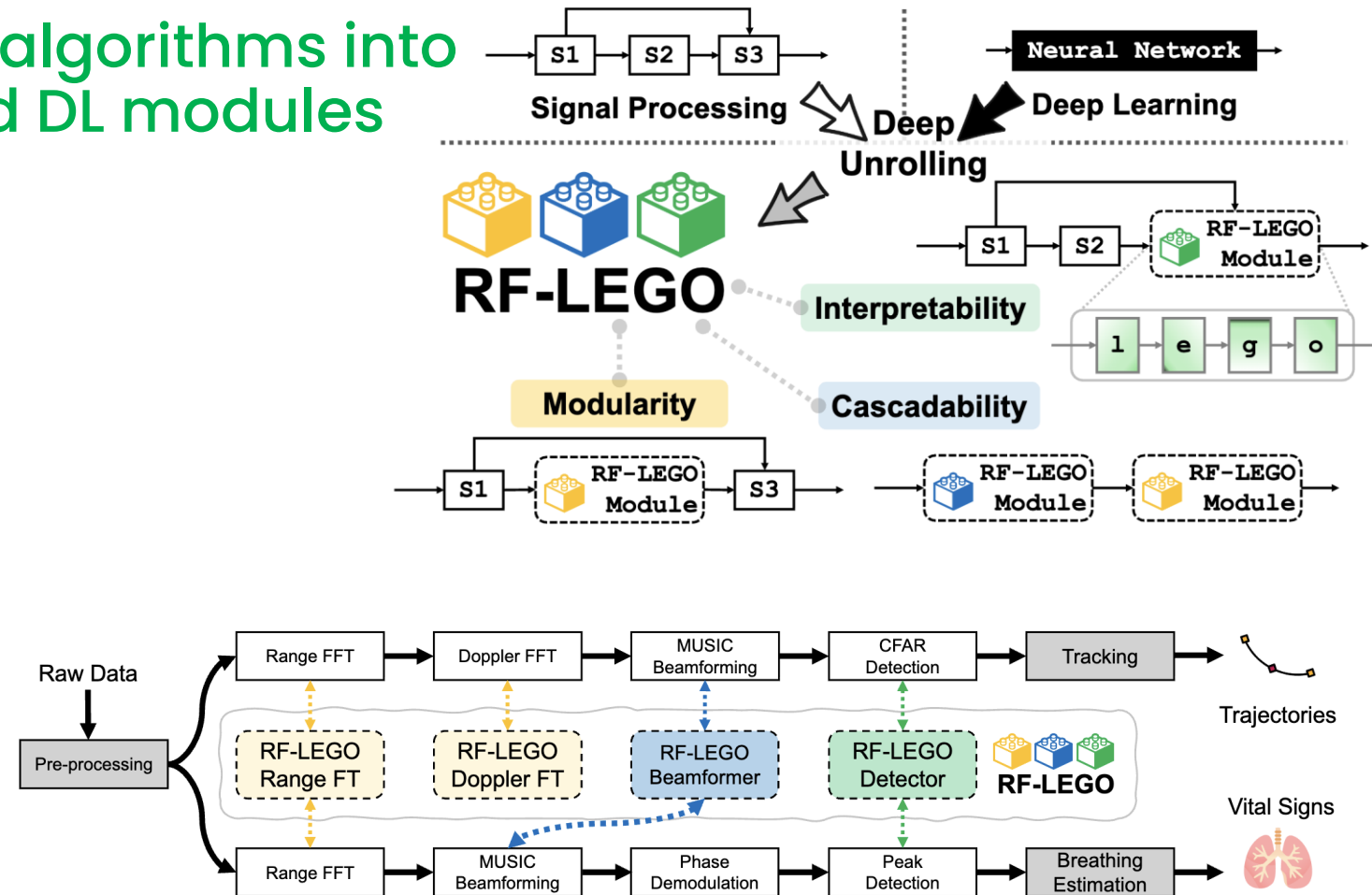


(c) The accuracy of breath rate estimation with raw and enhanced spectrograms.

RF-LEGO: Modularized SP-DL Co-Design via Deep Unrolling

- transforms interpretable SP algorithms into trainable, physics-grounded DL modules

- ✓ **Modularity:** Plug-and-play deep SP blocks
- ✓ **Cascadability:** Flexibly combined with each other or integrated into existing methods
- ✓ **Interpretability:** retain signal interpretability in both the processing structure and intermediate representations



RF-LEGO: Modularized SP-DL Co-Design via Deep Unrolling

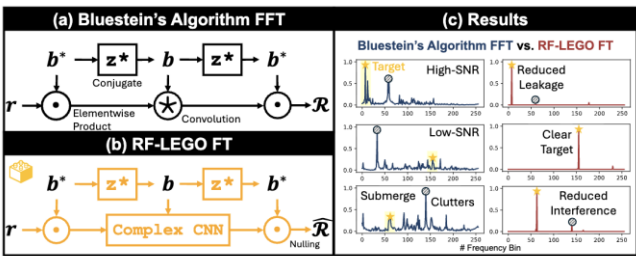


Figure 3: RF-LEGO FT.

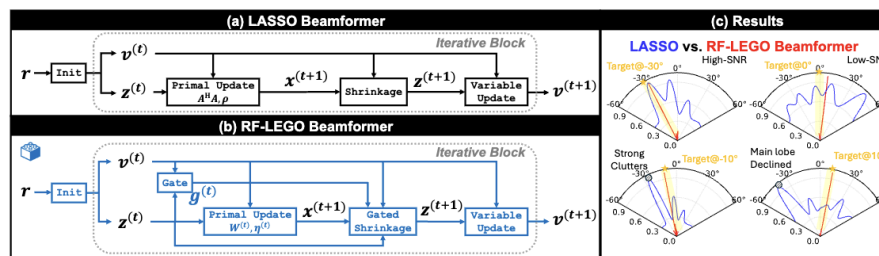


Figure 4: RF-LEGO Beamformer.

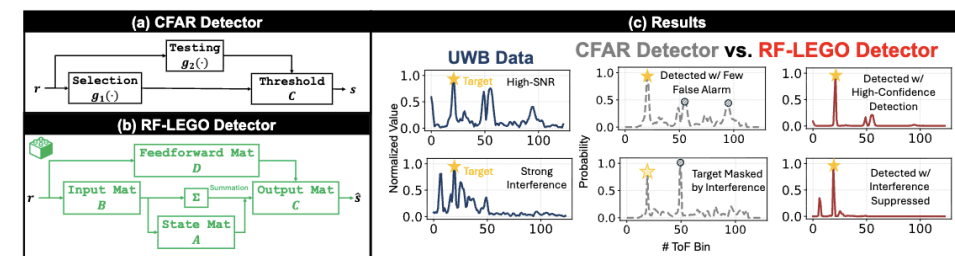
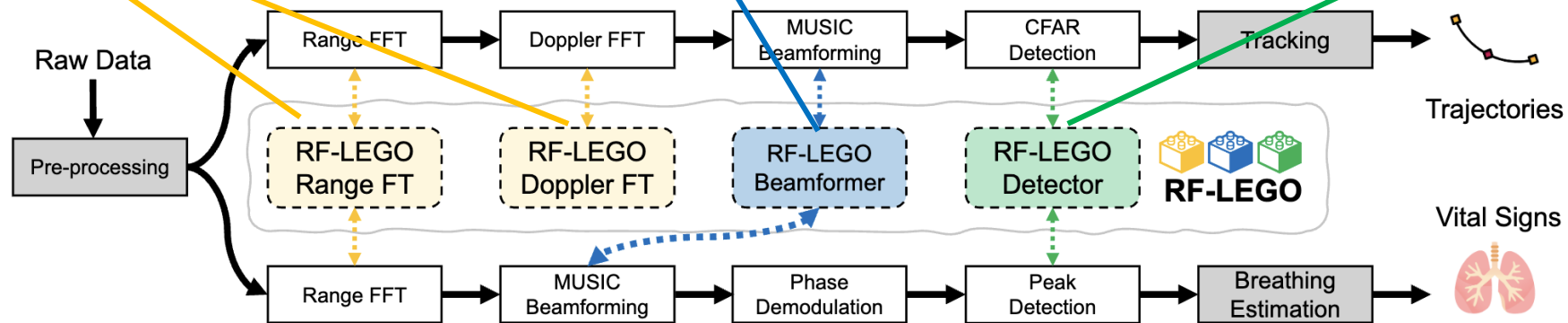
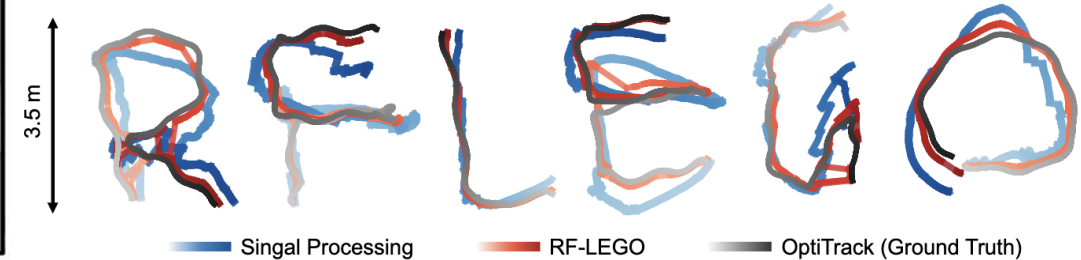
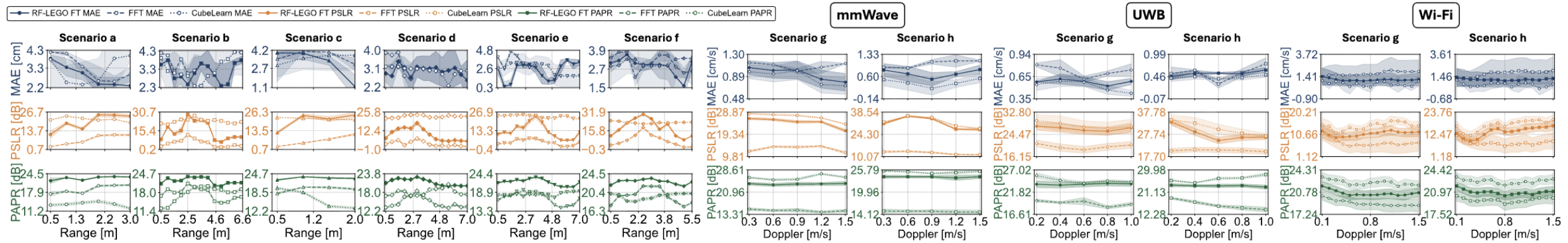


Figure 5: RF-LEGO Detector.



RF-LEGO: Modularized SP-DL Co-Design via Deep Unrolling



Luca Jiang-Tao Yu, Chenshu Wu, RF-LEGO: Modularized Signal Processing-Deep Learning Co-Design for RF Sensing via Deep Unrolling, MobiCom'26

RF-CRATE: White-box Deep Wireless Sensing

- Interpretable Deep Wireless Sensing via White-Box Transformer

- ✓ Fully mathematically interpretable deep network architecture for RF sensing
- ✓ Grounded in the principles of complex sparse rate reduction
- ✓ Complex-valued white-box transformer

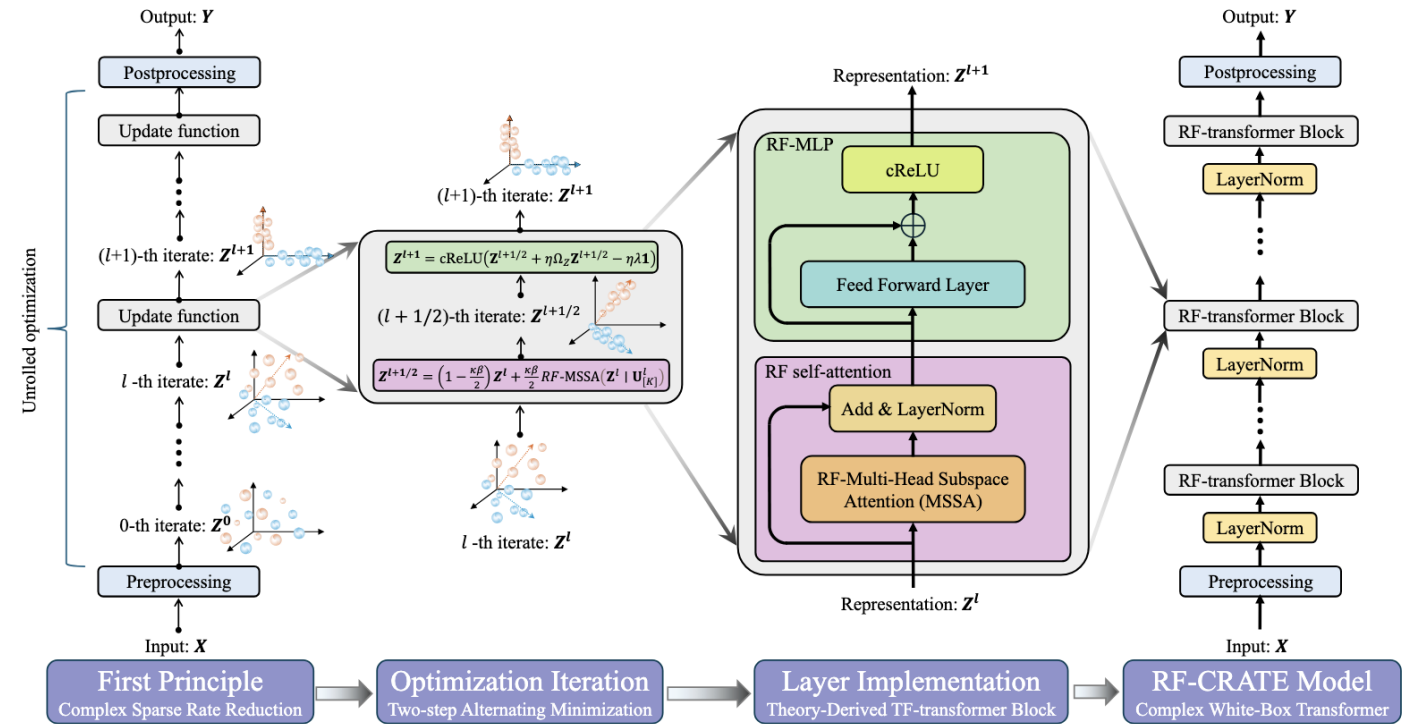
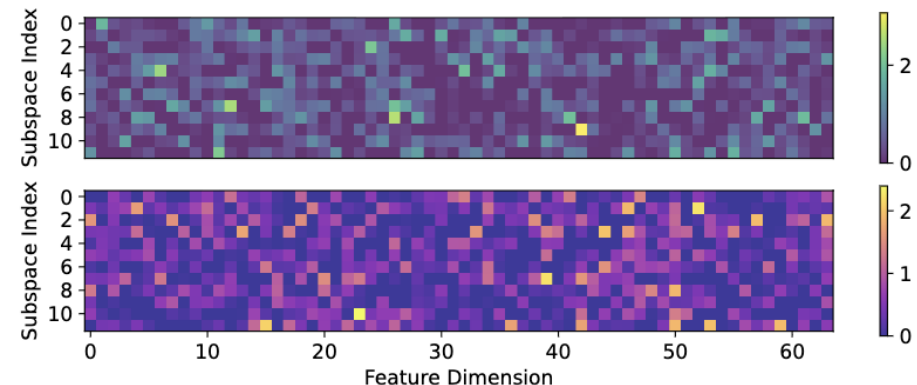
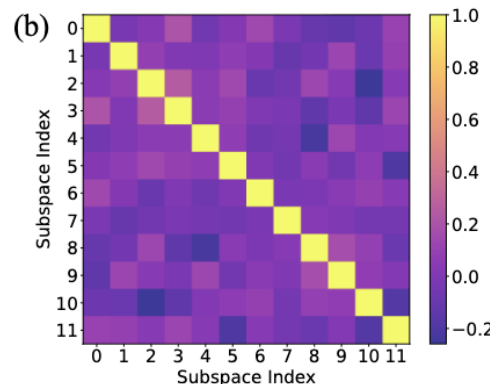
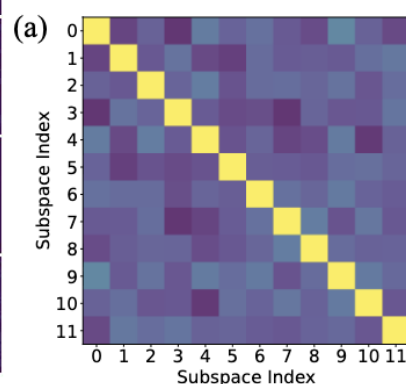


Fig. 2. Overview of RF-CRATE: From first principle to model structure.

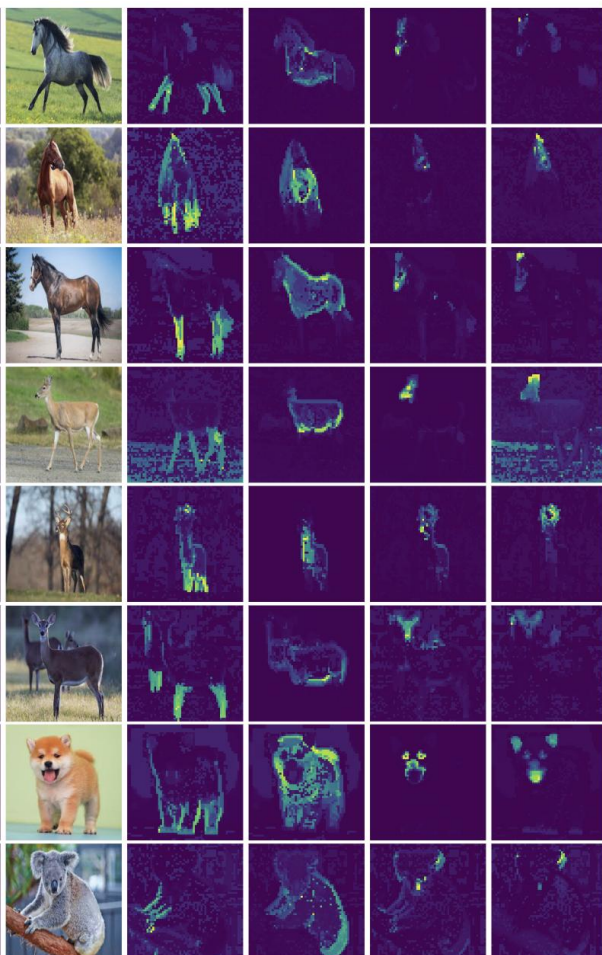
RF-CRATE: Complex White-box Transformer

Model Type	Models	WiFi Gesture (Acc. $\uparrow$)	WiFi Gait (Acc. $\uparrow$)	WiFi 3D Pose (MPJPE $\downarrow$)	WiFi Breath (BPME $\downarrow$)	mmWave Pose (MPJPE $\downarrow$)	UWB Activity (Acc. $\uparrow$)
Transformer	Swin-T	84.25 $\pm$ 9.96	96.86 $\pm$ 3.11	28.70 $\pm$ 26.23	2.44 $\pm$ 0.86	10.61 $\pm$ 6.02	67.02 $\pm$ 9.16
	Swin-T-V2	55.65 $\pm$ 18.45	92.70 $\pm$ 4.90	28.90 $\pm$ 25.99	2.44 $\pm$ 0.77	10.40 $\pm$ 6.54	62.61 $\pm$ 6.83
	CRATE	75.20 $\pm$ 12.92	98.96 $\pm$ 1.83	28.66 $\pm$ 26.28	2.44 $\pm$ 0.85	24.93 $\pm$ 11.96	57.92 $\pm$ 8.97
DWS	RF-Net	55.33 $\pm$ 15.91	99.00 $\pm$ 1.88	N/A	3.57 $\pm$ 0.98	8.37 $\pm$ 5.91	57.87 $\pm$ 5.28
	STFNet	50.61 $\pm$ 17.48	95.30 $\pm$ 4.14	N/A	N/A	47.20 $\pm$ 21.44	58.59 $\pm$ 7.59
	SLNet	73.67 $\pm$ 13.11	94.79 $\pm$ 4.14	N/A	N/A	N/A	N/A
	Widar3.0	50.45 $\pm$ 18.29	82.01 $\pm$ 6.53	N/A	N/A	62.40 $\pm$ 15.70	N/A
	RF-CRATE	82.10$\pm$10.41	98.98$\pm$1.91	28.12$\pm$27.02	2.44$\pm$0.83	17.09$\pm$9.13	60.07$\pm$10.41



Subspace correlation

Feature sparsity



Head 0 "Leg" Head 1 "Body" Head 3 "Face" Head 4 "Ear"

AM-FM: A Foundation Model for Ambient Intelligence Through WiFi

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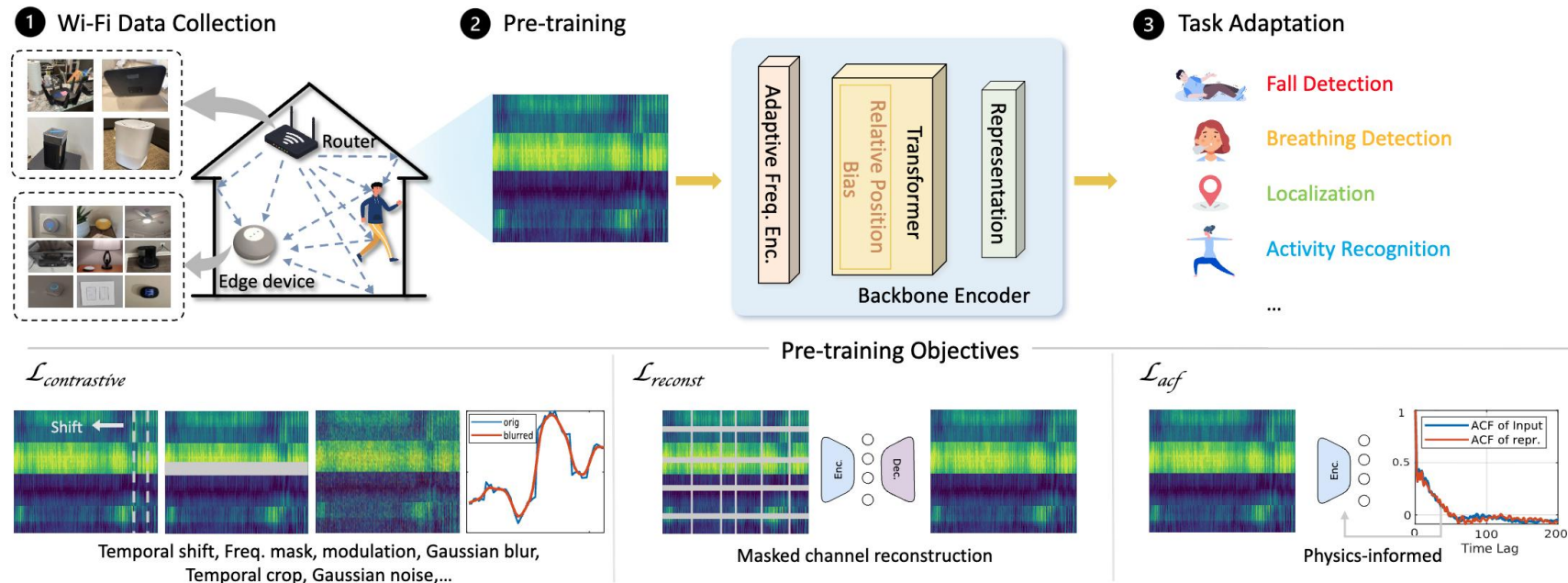


Figure 1. AM-FM Framework. (Left) Continuous CSI collection from distributed IoT devices. (Center) Self-supervised pre-training with contrastive learning, masked reconstruction, and physics-informed ACF prediction. (Right) Parameter-efficient adaptation to downstream tasks via lightweight temporal classifiers or bottleneck adapters.

DWS Models

- There are too many to list...
- Cross-domain generalizability is still a big issue
 - which is difficult to verify without a comprehensive dataset.
- Target to solve more challenging problems using DL, because well-addressed applications do not necessitate DL.

Questions?

- Thank you!